Immigration and Credit in America

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Abstract

We study the assimilation of immigrants into U.S. consumer credit markets. Although immigrants arrive without a U.S. credit history, we find that they are positively selected: immigrants' average credit scores by age thirty are 27 points higher than non-immigrants of the same age and five-digit ZIP code, and this difference persists with age. Despite greater creditworthiness, immigrants are less likely than non-immigrants to have ever had an auto loan or a mortgage by age 37, the end of our sample window. We compare credit access of same-age immigrants arriving one year apart, finding persistent differences in credit access lasting over a decade after immigration. Our results point to the importance of time in the U.S. for accessing credit.

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1. Introduction

When immigrants arrive in the United States, they bring their human capital and labor market credentials, which can help them navigate U.S. labor markets. However, immigrants start American life with a blank U.S. consumer credit report: any foreign credit history is generally invisible to U.S. credit bureaus and lenders. This missing credit history may impede access to credit, potentially undermining immigrants' ability to realize their significant economic potential in areas such as innovation and entrepreneurship (e.g., Azoulay et al., 2022, Bernstein et al., 2025).

This paper provides the first large-scale empirical evaluation of how immigrants assimilate into the U.S. consumer credit system. We use variation in the lifecycle timing of Social Security Number (SSN) assignment: individuals assigned SSNs as adults (at age 21+) are predominantly immigrants (Klopfer and Miller, 2024, Bernstein et al., 2025). We combine this fact with the sequential assignment of SSN blocks within states to classify immigrants and their immigration timing, in a 10% representative sample of U.S. consumer credit reporting data from TransUnion spanning 2000 to 2024. We study consumers born between 1975 and 1987 so that we can observe their entry into the credit system. Using this dataset, we track the progress of both non-immigrant and immigrant cohorts through the U.S. credit system, defining cohorts by their age at SSN assignment. This empirical strategy contrasts outcomes across immigration ages from 21 to 29 while including fixed effects for birth year and fine geography (five-digit ZIP code) to account for unobservables, enabling us to shed light on the connection between age at immigration and credit market trajectories.

Our core findings provide new empirical facts about immigration and credit in the United States. Once an immigrant enters the credit system, we find that they have 26.7 points higher credit scores by age 30, on average, compared to non-immigrants of the same age in the same geographic area. This average difference in credit scores persists as they age. Figure 1 shows that the distribution of credit scores by age 40 for immigrants is systematically higher than that of non-immigrants. Immigrants also have persistently lower delinquency rates and more conservative credit utilization patterns. This *positive selection* on credit quality is more pronounced among those who immigrate when they are older.[1]

However, despite immigrants' higher credit quality upon entering U.S. credit markets, which is

[1]Legal immigrants to the United States have a broad range of immigration statuses, including employment-based, immediate relative, and family-sponsored. Most legal immigrants to the United States are *not* H-1B visa holders (a type of visas for workers in specialty occupations, often requiring higher education). Official statistics from Office of Homeland Security Statistics Immigration Statistics (2024) show that from 2004 to 2023, only 16% of legal immigrant temporary workers admitted were on H-1B visas, and only 9% of legal new arrivals awarded lawful permanent residence (green cards) for employment are professionals with advanced degrees or exceptional ability.

captured by their credit scores, we find that they access less credit than non-immigrants. While the credit card access of immigrants converges relatively quickly to non-immigrant levels, significant differences persist in whether immigrants ever access auto loans and mortgages by the end of our sample frame. By age 37, immigrants are 15.7 percentage points (20.6% of the mean) less likely to have ever had an auto loan and 8.9 percentage points (20.0%) less likely to have had a mortgage than non-immigrants of the same age in the same geographic area. The immigrant mortgage difference is largely explained by immigration timing, with 1.8 percentage points lower access by age 37 for immigrants arriving each year later in their twenties. Even though *access* to credit cards exhibits fast and complete convergence to non-immigrant levels, credit card *limits* – an intensive margin of credit – lag behind, taking up to a decade to converge, which also appears largely due to the timing of immigration.

To assess whether these credit access gaps merely reflect differences in income or observable socioeconomic characteristics, we complement our credit reporting data analysis with public data from the American Community Survey (ACS). Using 5-year ACS waves that cover the years from 2006 to 2023, we show that immigrants arriving in their twenties have a lower likelihood of having a mortgage or owning a home, relative to non-immigrants in the same birth cohort, even after controlling for geography and detailed demographic characteristics. Despite immigrants having substantially lower household income (Appendix Figure A1), we also find that the mortgage and homeownership gaps persist after further controlling for these and other demographic factors, suggesting that lower credit access is not attributable to income differences.

A key mechanism behind these results is that the early timing of an immigrant's financial experiences sets a trajectory for their future—a phenomenon we term *history dependence*. To evaluate this idea, we use a paired cohort strategy that compares immigrant cohorts born in the same year, but who immigrated a single year apart. This strategy isolates the effect of a single additional year in the U.S. on long-term access to credit. We find that immigrants arriving to the U.S. a year later than same-age control immigrants have lower average access to auto loans and mortgages. Furthermore, this reduced credit access persists more than a decade post-immigration. The fact that immigrating a year later in your twenties matters for auto loans and mortgages, which are often accessed in a consumer's thirties, suggests that delayed entry into credit markets has long-term impacts on credit access.

We consider alternative mechanisms. Immigrants are more likely to leave the U.S. within our sample period, but we show that this emigration does not explain our results. Alternatively, immigrants may have a lower demand for credit, but we find that they have greater demand for credit in their twenties, both relative to non-immigrants and relative to same-age immigrants arriving a year before them. Further, using the Survey of Consumer Finances (SCF) we find that immigrants have a more negative attitude towards auto debt than non-immigrants, but no greater general aversion to debt (e.g., Martínez-Marquina and Shi, 2024), suggesting that this cannot explain our results for credit card and mortgage use.

Our paper contributes to the understanding of immigrants, their entry into the U.S. credit system, and the importance of credit history for credit access in the U.S. First, our findings provide a new perspective on how immigrants are selected, a major theme in the immigration literature. As reviewed in Abramitzky and Boustan (2017), immigrants to the U.S. have swung between being positively selected and negatively selected, depending on the immigration wave, measure of selection, and the country of immigration. We contribute to this literature in two ways. First, we provide a characterization of immigrants via detailed consumer credit reporting data. Relative to labor outcomes, such as education and income, credit reporting data offer a complementary perspective on immigrant characteristics and selection. For example, we observe delinquency rates and the usage of all major types of credit. Second, our analysis provides a representative picture of how immigrants compare to non-immigrant consumers in terms of their creditworthiness. Creditworthiness is a different and important characteristic from the primarily labor market outcomes studied in the immigration literature. Income and other labor market characteristics (e.g., education, employment) are not inputs to credit scores (Gibbs et al., 2025) and credit scores have little correlation with income (Berman et al., 2026). Indeed, Chatterjee et al. (2023) consider that a credit score is a market assessment of a person's unobservable type, which they interpret as patience.

In addition, our findings provide a fresh perspective on the assimilation of immigrants into the U.S. economy. There is a robust literature on the assimilation of immigrants, which has characterized the labor market and cultural assimilation and social integration of immigrants during different immigration waves (e.g., Borjas, 1985, Abramitzky et al., 2014, 2020, 2021, Lubotsky, 2007, Bleakley and Chin, 2010, Bailey et al., 2025, Doran et al., 2022, Bazzi and Fiszbein, 2025, Kerr et al., 2026). Our evaluation of assimilation into U.S. consumer credit markets not only offers a distinct venue of assimilation into the U.S. economy — credit markets versus labor markets — but it also suggests that the reliance on credit history in credit markets, and the inability to port credit histories across national boundaries, can slow assimilation into the credit system, leaving a lasting impression on immigrant credit access. While much of the immigration literature focuses on labor market assimilation, our evidence from both credit reporting data and the ACS shows that asset-linked credit outcomes — homeownership and mortgage use in particular — exhibit persistent gaps that are not fully explained by observable income differences.

We also contribute to the household finance literature on frictions in credit scoring, credit access and real effects. By studying immigrants—an important consumer segment with shorter credit profiles due to their later credit market entry—our research provides a new perspective on credit access for consumers with "thin" credit reports that only contain a few accounts or a short credit history (e.g., Di Maggio and Ratnadiwakara, 2026, Blattner and Nelson, 2024) or no credit history (e.g., Chioda et al., 2025).[2] For example, it is striking that immigrants' delayed entry into U.S. credit markets delays mortgage and auto loan access by age 40, even though immigrant credit scores are higher than those of non-immigrants.

Our findings on delayed entry into the U.S. credit system also relate to the work on the long-term impacts of delayed credit market entry (Brown et al., 2019). Related work has emphasized the importance of a good start in credit markets, either through good parental credit histories (e.g., Bach et al., 2023, Benetton et al., 2025, Bakker et al., 2025) or starting one's credit history in good economic times (Ricks and Sandler, 2025). Relative to this literature, our variation in credit market entry timing (and thus history) deploys a person-specific, near-mechanical reason for delay — immigrants cannot enter U.S. credit markets before immigration, nor get credit for their credit history in other countries. Although our results on credit access are similar to other factors that delay and restrict access to credit, our findings on immigrants' higher credit scores contrast with work on credit entry timing and later credit scores (Nathe, 2021), suggesting that later credit market entry has a smaller impact on credit scores for immigrants.[3]

The paper proceeds as follows. Section 2 explains our data, how we classify immigrants, institutional details of lending to immigrants, and our empirical strategy. Section 3 shows our results. Section 4 contains the methodology and results for the paired cohorts analysis. Section 5 evaluates the mechanisms behind our results and Section 6 concludes.

[2] Our perspective on immigrants differs in at least two ways from recent household finance research on immigrants and the dynamics of credit via movers. First, as reviewed in Gomes et al. (2021), the existing household finance literature on immigration has largely focused on cultural differences (e.g., Carroll et al., 1994, Osili and Paulson, 2004, Haliassos et al., 2017, Fuchs-Schündeln et al., 2020, Zillessen, 2022, Gorback and Schubert, 2025). Second, we study the credit access of immigrants, which is a different focus to prior work by Keys et al. (2023) that has studied how consumer financial distress varies after moves within the United States.

[3] Our paper complements Kovrijnykh et al. (2024), which shows that consumers can boost their overall access to credit not only via prompt repayment, but also via by opening new credit cards.

2. Data and Empirical Design

In this section, we describe the data that we use and explain how we generate the two key datapoints we need for each consumer: whether they are immigrants and, if so, when they immigrated. Subsection 2.3 provides context on lending to immigrants in the United States, while subsection 2.4 explains our empirical methodology.

2.1 Data

2.1.1 Consumer Credit Reporting Data

We use consumer credit reporting data from the University of Chicago Booth TransUnion Consumer Credit Panel, "BTCCP" (TransUnion, 2025). The BTCCP is an anonymous, representative sample of U.S. consumer credit reporting data provided by TransUnion that contains monthly information from July 2000 to December 2024. To build this sample, TransUnion starts with a 10% representative sample of consumers who had a credit report in July 2000. For each month of data after July 2000, 10% of new consumers entering the main TransUnion data are added to the BTCCP to maintain its representativeness. We follow the best practices for using consumer credit reporting data described in Gibbs et al. (2025).

The BTCCP is a collection of datasets, linkable to one another via a consumer identifier. We primarily construct our outcomes from the BTCCP's monthly tradeline-level dataset, which contains information (i.e., outstanding balance and delinquency status) on each credit account held by a consumer. This tradeline dataset provides the date each account was opened, including those opened or closed prior to July 2000. We also use the BTCCP's consumer-level header dataset, which contains each consumer's birth date and the date a consumer first has a credit report, even if this was before July 2000. By combining the consumer's birth date with the account opening date in tradeline data, we compute the consumer's age at account opening.[4] From the BTCCP's monthly consumer-level aggregated dataset we use a consumer's credit score, VantageScore, their ZIP code and state, and months since the last credit inquiry (search). From this last dataset, we use an annual panel of aggregated credit characteristics (observed each July) to construct outcomes, and use monthly data to calculate control variables for consumers' first credit score, first ZIP code,

[4] We follow standard practice in removing tradelines that have not been updated in the last twelve months (Gibbs et al., 2025). Some tradelines appear to be opened before the age of 18. We drop such cases to produce more plausible estimates of credit usage at age 18, as these cases are before a consumer can take out a credit agreement by themselves, and are likely to either be a data error or a credit product taken out by their parent.

longest ZIP code, and the number of unique ZIP codes.

2.1.2 American Community Survey (ACS)

We use public Census data from the 5-year American Community Survey (ACS), accessed via the Integrated Public Use Microdataset Series (Ruggles et al., 2025). These anonymous data contain birth years, years of immigration, geographic location, and detailed socio-economic information. We use the 2010 wave as a benchmark to evaluate our immigrant classification. We also use 5-year repeated cross-sectional waves that cover the years from 2006 to 2023 to evaluate the mechanisms behind our results.

2.1.3 Survey of Consumer Finances (SCF)

We use public data from the nationally-representative cross-sectional Survey of Consumer Finances, SCF, (Board of Governors of the Federal Reserve System, 2023) to help evaluate the mechanisms behind our results. We use the 2022 wave of the SCF as this is the only wave that includes information on whether a consumer is an immigrant and the number of years since immigration. The SCF contains detailed socio-economic information, however, does not contain geographic information or credit scores.

2.2 Immigrant Classification

We classify whether a consumer is an immigrant and their age at immigration following the "sequential SSN assignment" procedure used in recent research on U.S. immigration (Yonker, 2017, Doran et al., 2022, Bernstein et al., 2025, Klopfer and Miller, 2024, Engelberg et al., 2025).[5] SSNs are nine-digit numbers in use since 1936 that are unique to a consumer and are assigned by the Social Security Administration (SSA). Up until mid-2011, the SSA sequentially assigned SSNs using the same first five digits before moving onto another five-digit block, with the timing of these allocations being public. The first three digits denote an "area number," a geographic region within a state, the fourth and fifth digits are the "group number," assigned sequentially in blocks within that area, and the last four digits are the order of processing, which is quasi-randomly assigned (see

[5] Advani et al. (2025) also uses a related method to study immigration in the United Kingdom. The closest prior data to ours is Federal Reserve (2007) report to Congress, which evaluated traditional credit scoring models in a 2003 sample of 300,000 credit records that included immigrants. Our dataset offers a substantial advancement over this early work because it is more recent, more comprehensive, more granular, and has information on immigration age. Because of these features and because our data cover 25 years, we are able to speak to the question of assimilation into U.S. credit markets by evaluating the dynamics of immigrant credit.

Puckett, 2009 for a comprehensive history of SSN assignment). As established in prior research, consumers assigned an SSN at age 21 or later in life are generally immigrants (Klopfer and Miller, 2024, Bernstein et al., 2025).

TransUnion observes SSNs, however, the BTCCP that we have access to is anonymized without SSNs, so we developed a procedure to enable us to classify immigrants in our data without directly accessing SSNs. To estimate the year of SSN assignment, $SSN\ Year$, we construct a lookup table using public information that maps the first five digits of the SSN into the first year of SSN assignment, $SSN\ Year$, for all SSNs assigned before 2012. We sent TransUnion a list of anonymous consumer identifiers in the BTCCP, together with this lookup table. TransUnion then matched the consumer list and the lookup table to their underlying data, which includes consumers' SSNs, returning a dataset with the year of SSN assignment, an indicator for whether a consumer had any SSN in their data, and the BTCCP consumer identifier. This procedure guarantees that we neither observe nor can infer SSNs for any consumers in the BTCCP. We follow the prior literature in classifying consumers as an immigrant if their age at SSN assignment, $SSN\ Age$, is greater than or equal to 21, where $SSN\ Age = SSN\ Year - Birth\ Year$.

For our analysis, we apply two sample restrictions. First, we restrict to consumers who have an SSN in TransUnion and whose birth years are between 1975 and 1987. Consumers with these birth years are mostly expected to enter the credit system after 2000 (and in our data we observe their accounts that were opened before 2000) and before the SSN assignment period ends in 2011. For example, from 2000 to 2024 (the BTCCP coverage window), consumers born in 1975 are observed from ages 25 to 49, while consumers born in 1987 are observed from ages 18 to 37. This part of the lifecycle – early adulthood to middle age – is when we expect credit access to be most economically important.[6] This follows similar approaches used in related papers (Ricks and Sandler, 2025, Bach et al., 2023, Bakker et al., 2025). Restricting to birth years 1975 to 1987 also corresponds to the subset of birth years in our data where the estimated immigration shares closely resemble those in the ACS, as shown in Panel A of Appendix Figure A2. Second, we restrict our attention to consumers with $0 \leq SSN\ Age < 30$, which focuses our analysis on immigrants arriving in the United States at some point in their twenties. This ensures that, for all consumers, credit outcomes in a consumer's thirties occur *after* everyone in our sample has immigrated and we can examine longer-term credit outcomes observed by 2025. Additional details on our data construction are in

[6]For some analyses that rely on outcomes constructed from the BTCCP's consumer-level datasets, we further restrict to birth years 1982 to 1987 to ensure that consumers enter the credit system after 1999.

Appendix A.1.

After applying these restrictions, we have a total of 6,299,287 consumers, 6.10% (384,396) of whom we classify as immigrants (i.e., SSN Ages 21 through 29). Appendix Figure A3 shows that this sample is consistent with the ACS data. Although some older consumers were assigned SSNs as adults, for the birth years that we study, non-immigrants were largely assigned an SSN at birth or before turning 18, as demonstrated in administrative Social Security Administration data by Klopfer and Miller (2024).[7] The consumers we classify as immigrants are more geographically mobile than non-immigrants (Appendix Figure A5).

To help evaluate the external validity of our data restrictions, we examine the 2010 5-year ACS data. We find a similar distribution of income and education levels in the ACS with and without imposing our sample restrictions. Compared to non-immigrants, immigrants typically have lower incomes and are generally less educated. Immigrants are more likely than non-immigrants to be educated at the doctoral level, however, this represents only 2.1 percent of immigrants. See Appendix Figure A1 for details.

Our classification of immigrants excludes certain groups. First, our immigrant classification does not include illegal immigrants (who lack SSNs), and who are less likely to use the formal credit system for demand and supply reasons. Second, we do not observe immigrants or non-immigrants who remain entirely outside the formal credit system, because we study formal credit only, not informal credit. Third, we do not study immigrants who arrive before adulthood or at older ages. Fourth, we do not include legal immigrants with Individual Taxpayer Identification Numbers (ITINs) or Enumeration at Entry numbers (EAE), as we do not know when these are assigned. However, these are typically only used before a consumer receives an SSN.

2.3 Institutional Details on Lending to Immigrants

Lenders in the United States may consider an applicant's immigration status in their lending decisions – immigration status is *not* a protected characteristic under the Equal Credit Opportunity Act (ECOA). This contrasts with ECOA-protected characteristics such as gender, national origin, and race. Immigration status is observed in auto loan and mortgage lending applications as a by-product of requesting documents (e.g., passport) to verify the identity of an applicant. For

[7] If we instead classify immigrants as those with SSN ages greater than 18 the comparison is unaffected, as the non-immigrant baseline is almost unchanged and only 109,207 consumers have SSN Ages of 19 or 20 – see Appendix Table A1, Panel B. Appendix Figure A4 shows the distribution of SSN Ages by birth years in our data, which mirrors that in Klopfer and Miller (2024).

credit card applications, a lender may only know whether an applicant has an ITIN or an SSN. As detailed in Consumer Financial Protection Bureau (2023), lenders have been constrained in how much they can rely on immigration status in their underwriting as they need to avoid discriminating on ECOA-protected characteristics, however, visa status may limit which types of mortgages an immigrant is eligible for.[8]

Lenders in the United States use credit scores and other information in their lending decisions. Information frictions may affect lenders' decisions and impact immigrants' access to credit in several ways. First, immigrants initially lack a U.S. credit report and a credit score. Lenders may therefore have a prior that, given no information, immigrants are high credit risk. Second, after entering the U.S. credit system, immigrants have a shorter length of credit history than most non-immigrants of the same age, who often enter the credit system at age 18 or in their early twenties. The length of credit history positively affects credit scores (FICO, VantageScore) and may also be a direct input in lenders' underwriting via overlays applied in combination with scores (e.g., applicants need at least two years of credit history irrespective of score) or as an input into proprietary scores that are created by lenders (Gibbs et al., 2025).

Consumer credit markets vary in their reliance on credit scores and on information in a credit report (including the length of credit history), with credit card decisions typically being more reliant on credit scores than auto loan or mortgage decisions (Blattner et al., 2022). Credit card lending decisions are primarily based on scores (e.g., Agarwal et al., 2018). Whereas for auto loans, the lender may evaluate other information, such as income, especially if a consumer does not have a very high credit score (e.g., Grunewald et al., 2025). For mortgages, Fannie Mae and Freddie Mac mortgage eligibility requirements depend not only on credit scores, but also on information obtained directly from the borrower, such as loan-to-value and debt-to-income ratios calculated using verified income information, and the liquid assets available to a consumer after closing.[9]

The length of credit history can also have indirect impacts on credit access, as, in the U.S. credit system, consumers need to take out credit to access credit. Less time in the credit system means

[8] A change in government policy towards immigrants may be altering this. In March 2025, the U.S. Department of Housing and Urban Development (2025) removed eligibility for "non-permanent residents" (legal immigrants) from Federal Housing Administration programs (illegal immigrants were always ineligible). This policy change is too recent to affect our results, but may reduce immigrants' access to mortgages. Similarly, financing auto loans to immigrants appears to be becoming harder in 2025. One auto lender, Tricolor, which specialized in selling and financing used cars to immigrants, filed for bankruptcy in 2025 (although this may reflect fraud). Another subprime auto lender, LendBuzz, warned of increased risks of lending to immigrants in its 2025 filing ahead of its Initial Public Offering.

[9] Other factors could also lead lenders to restrict credit to immigrants. For example, in the United Kingdom, Dobbie et al. (2021) show how one subprime consumer lender's decisions are biased against immigrants due to loan examiner incentives focused on short-run outcomes, rather than long-run profits.

less time to grow the number of credit accounts on a consumer's credit report. This can impede credit access, as it makes immigrants more likely to have "thin files", where they hold only a few credit accounts, which results in noisier assessments of credit risk (Blattner and Nelson, 2024). A shorter credit history can also mean less time for credit card limits to grow. This can reduce credit access because lenders increase card limits over time and they also are influenced by the existing limits assigned to a consumer by competing lenders (De Giorgi et al., 2023, Kovrijnykh et al., 2024).

Cultural barriers may also prevent immigrants from accessing credit, such as differences in familiarity and willingness to use or repay debt (e.g., Bursztyn et al., 2019). Language barriers may play a role (Liu, 2025), and immigrants may not understand the U.S. credit system, particularly how one needs to take out and use credit to build and improve their credit score.

2.4 Empirical Methodology

We use two main empirical specifications to analyze outcomes in our credit reporting data. For both of these specifications, we collapse the data to one observation per consumer i estimating OLS cross-sectional regressions. Our first main specification is Equation (1):

$$\mathrm{Y}_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i, \tag{1}$$

In this specification, β_1 is the coefficient of interest, estimating the average difference in outcomes between immigrants and non-immigrants. This is the coefficient on an indicator variable, $\mathbb{I}\{Immigrant\}_i$, which takes a value of one for *immigrants* – consumers with SSNs assigned at ages 21 through 29 – and takes a value of zero if their age of SSN assignment is less than 21 (we classify these as *non-immigrants*). This regression also includes birth year fixed effects ($\mu_{b(i)}$), as well as geographic fixed effects ($\eta_{g(i)}$) for the first five-digit ZIP code (First ZIP5) associated with a consumer in our data, to capture unobservable heterogeneity in local market conditions.

Our second main equation is Equation (2):

$$\mathrm{Y}_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i, \tag{2}$$

In this specification, we include a linear term for the age of SSN assignment ($SSN\ Age_i$) to allow for heterogeneity in immigrant outcomes. To aid interpretation, we pool all non-immigrant SSN Ages (0 through 20) into one category. The β_2 coefficient reflects the marginal difference in average outcomes associated with being assigned an SSN one year later in life. In both of these specifications, we cluster standard errors by birth year $\times$ the first two digits of a consumer's first observed ZIP code (a coarser unit of geography than ZIP5 that corresponds to state or subregion).[10]

In the Appendix, we show that our results are robust to adding fixed effects for additional geographic controls: the most recent five-digit ZIP observed for a consumer in our data (*Last ZIP5*), the five-digit ZIP observed for a consumer for the most months in our data (*Longest ZIP5*) which is a measure of their most permanent location, and the number of unique five-digit ZIP observed for a consumer (*Number ZIP5*) which is a measure of their geographic mobility.[11] All of these geographic controls are imperfect because consumers enter and exit our data at different times, partially based on their credit access. In the Appendix, we also show that our results are robust to restricting to consumers who have not emigrated, which we measure by conditioning on the subsample of consumers who are still observed in the data in 2024.

An important implementation detail is how to handle missing values of cross-sectional outcome variables, such as the age at which a consumer first takes out a mortgage. Because these consumers never accessed the credit product, excluding these from the estimation would understate any difference in credit access. We address this issue by assigning these observations a value that corresponds to accessing the product in 2025, one year after the end of our data.

3. Results

This section describes the main results on immigrant credit entry, creditworthiness and credit access. We begin with summary statistics to examine cross-sectional differences in immigrant and non-immigrant credit, then move to our main analysis, with an alternative empirical design presented later in Section 4. In Section 5, we then discuss the mechanisms that could explain our results.

3.1 Summary Statistics

Table 1 presents sample means of variables for immigrants ($SSN\ Age$ 21+) versus non-immigrant consumers ($SSN\ Age\ <\ 20$). These statistics highlight first-order differences in the nature and timing of immigrant versus non-immigrant consumer credit use.

[10]Because our sample is large, statistical significance at conventional levels is attained in almost all of the tests we run; the statistical significance of our estimates is similar if we cluster by birth year.

[11]Geographies vary in their economic opportunities or access to finance, and this may disproportionately affect immigrants. That is, immigrants may be more likely to arrive in cities rather than rural areas, or in areas with different incomes or other economic opportunities, and with consequently different credit access or economic vibrancy. This motivates our granular geographic controls.

Immigrants typically enter the U.S. credit system later than non-immigrants: The average age at first credit report for immigrants is 26.1 compared with 20.2 for non-immigrants. Immigrants' later age in accessing their first credit card is similar, with immigrants having an average age at first credit card of 27.2 versus 22.3 for non-immigrants. Though delayed, immigrants are just as likely to eventually have a credit card within our sample frame (97.4% versus 96.9% for non-immigrants).

Creditworthiness can be evaluated by the average credit score (VantageScore, where a higher score is higher credit quality), which is *substantially* higher for immigrants than for non-immigrants. By age 30, the average credit score of immigrants is 658.3 compared to 624.2 for non-immigrants, a 34.1 point difference. Some of this difference could be explained by immigrants of higher credit quality selecting into having credit scores at earlier ages, with immigrants being 14.6 percentage points less likely to have been scored by age 30. However, the difference between immigrants and non-immigrants in average credit scores persists by age 40 with immigrants having 29.4 points higher scores than non-immigrants, and by that point in the lifecycle effectively all consumers in the sample have been scored.

Turning to credit products typically accessed later in a consumer's lifecycle, we find that immigrants are significantly less likely than non-immigrants to access auto loans (65.6% for immigrants versus 81.1% for non-immigrants) and mortgages (44.4% versus 50.3%). Conditional on consumers who access these loans in the sample, immigrants access both auto loans and mortgages later than non-immigrants. Their average age at first auto loan is 30.6 (versus 26.1) and their average age at first mortgage is 32.9 (versus 29.5).

We also highlight differences in credit limits between immigrants and non-immigrants. At age 30, immigrants have similar, but slightly lower average credit limits than non-immigrants ($10,211 for immigrants versus $11,492 for non-immigrants), but by age 40, immigrants' total credit limits are $4,496 higher on average ($26,608 versus $22,112). In the following subsections, we will examine each of these differences and dynamics more precisely, conditioning on the age of SSN assignment, and accounting for birth year and geography with fixed effects.

3.2 Credit Market Entry

We first present evidence that the age at which a consumer receives an SSN (SSN Age) drives the timing of credit market entry, as expected. Most directly, our evidence shows that later in life SSN Age strongly predicts later credit market entry, both for the consumer's first credit product and for the first date on which they receive a credit score. These findings primarily validate SSN Age as a proxy for immigration timing.

Table 2 presents the results from estimating our regression specifications (equations (1) and (2)). Columns 1 and 3 show that immigrants are 5.5 years older than non-immigrants when they receive their first credit report, and 5.2 years older when they receive their first credit product, after controlling for birth year and first ZIP5 via fixed effects. Delayed entry into the credit system is more pronounced for immigrants who arrive in the United States later in life. When we add SSN Age to the specification (in columns 2 and 4) the mean delay relative to non-immigrants falls to 2.3 and 2.1 years, for first credit report and product, respectively. For each additional year of later in life SSN assignment, the first credit report (product) is delayed by an additional 0.75 (0.73) years. We interpret this delay as primarily mechanical: immigrants who would have counterfactually received a credit report or product before their SSN age could not do so – they were outside the country.[12] These results are also robust to adding fixed effects for the consumer's *last* ZIP5 and *longest* ZIP5, and *number* of unique ZIP5s, as shown in Appendix Table A2. These delays are economically significant relative to the average age that non-immigrants receive their first credit report (aged 20.2) and their first credit product (aged 21.6). Panel A of Appendix Figure A6 shows an essentially linear increase in the average age at first credit product for each additional year of SSN Age. This linearity reflects the reality that immigrants could not obtain credit products in America in the years prior to their immigration.

3.3 Creditworthiness

A core aim of the immigration literature is to understand how immigrants assimilate into the U.S. economy (e.g., Borjas, 1985). Our data enable us to measure how a consumer's creditworthiness and credit access evolves over their lifecycle via two comparisons: comparing non-immigrants to immigrants, and comparing the outcomes of immigrants assigned SSNs at different ages. In this section, we provide evidence on selection of immigrants, and then in the two following sections we study how immigrants assimilate into U.S. credit markets.

[12]One might expect that each additional year of SSN Age would mechanically delay the consumer's entry by one year. However, the estimates of around 0.75 are relative to non-immigrants, not all of whom have entered the credit system at each age, as displayed in Panels A and B of Appendix Figure A7. This pushes the mechanical effect below one.

3.3.1 First Credit Score

To complement the cross sectional regressions, in Figure 2 and later analyses that follow, we visually summarize our data for each combination of *SSN Age* and *Age* observed in our panel data, grouping consumers assigned an SSN at age 18 or earlier into a single benchmark category of non-immigrants with black lines. Other SSN Ages contain immigrants and have shades of blue lines, with lighter shades for consumers assigned SSN Ages later in life. This figure, and subsequent analogous figures for other outcomes, show SSN Ages 19 and 20 separately, to illustrate how classifying these consumers as either immigrants or non-immigrants would not impact our conclusions (consistent with Bernstein et al., 2025, whose results are not sensitive to whether immigrants are classified based on SSN Ages 19+, 20+, or 21+).

We examine the timing of consumers' first credit scores. Panel A of Figure 2 displays the unconditional likelihood of having a credit score for each SSN Age cohort at different ages. At age 18, 14% of the SSN Age 18 cohort of non-immigrants had received their first credit score; by age 22, 84% of these consumers have a score. The consistent rightward shift in these lifecycle curves as SSN Age increases indicates that consumers with later in life SSN Ages also receive a score later in life. In the years prior to the SSN Age (where $Age = SSNAge$ is indicated by a circle on each lifecycle curve), the likelihood of having a credit score is very low, and this is followed by a sharp jump in the percentage of scored consumers in the year following immigration. A small share of consumers have a credit score slightly before they receive an SSN because it is possible to access credit without an SSN (e.g., some consumers will have an ITIN before receiving an SSN).

Shortly after their immigration year, immigrants (i.e., those with SSN Ages 21 to 29), quickly converge to the SSN Age 18 non-immigrant cohort (the black line in the figure) in their likelihood of having a credit score. For example, while only 9% of SSN Age 22 consumers have a credit score by age 22, 75% have a credit score by age 26. The likelihood for immigrants assigned SSNs later in life converges more quickly. Moreover, immigration cohorts arriving later in life have a higher likelihood of having a credit score in their year of immigration. For example, only 8% of 21 year old immigrants (SSN Age of 21) have a credit score by age 21 whereas 22% of 29-year-old immigrants have a credit score by age 29. Because SSN Age 19 and 20 cohorts likely contain a mixture of immigrants and non-immigrants, these two cohorts fall between the SSN Age 18 and SSN Age 21 cohorts.

3.3.2 Mean Credit Scores

Panel B of Figure 2 plots the mean credit score for each SSN Age cohort at different ages, *conditional* on having a credit score at that age. It is important to study credit scores at different ages because a consumer's borrowing needs and propensity to repay debt can change over the course of their lifecycle (see Chatterjee et al., 2023, for a theory of credit scoring). Strikingly, this plot shows that immigrant SSN Age cohorts (SSN Age 21+) start out with a higher mean credit score than the (non-immigrant) SSN Age 18 cohort has at the same age. Immigrant cohorts continue to have substantially higher average credit scores throughout their thirties. Although the high average credit score in the year of immigration may partly reflect selection (i.e., immigrants who are scored are of higher credit quality), immigrant cohorts have higher credit scores than the SSN Age 18 cohort even once a similar fraction of both the immigrant cohorts and the SSN Age 18 cohort are scored. This suggests that immigrants who are scored are more creditworthy than their scored non-immigrant counterparts of the same age, on average. Overall, these dynamics paint a somewhat unexpected picture: Even new immigrants with a credit score are *observably* more creditworthy.

To provide a more formal characterization of these lifecycle patterns, in Panel A of Table 3 we report estimates from equations (1) and (2) for how average credit scores by ages 30 and 40 depend on immigration status and SSN Age. These regressions control for first geographic location (ZIP5) and birth year, neither of which are inputs to credit scores (Gibbs et al., 2025). Consistent with the descriptive lifecycle plots, column 1 shows that immigrants have average credit scores by age 30 that are 26.7 points higher than non-immigrants. In column 2, we allow for heterogeneity by SSN Age, and find that the mean difference in credit score by age 30 between immigrants and non-immigrants is 17.2 points, and increases by an additional 2.3 points per year of SSN Age.

The difference in average credit scores between immigrant and non-immigrants persists as consumers age. Column 3 of Panel A of Table 3 shows that immigrants' average credit score is 24.6 points higher at age 40. Column 4 accounts for heterogeneity by SSN Age, showing that the mean difference is 14.2 points and increases by an additional 2.4 points per year for consumers who immigrate later in life. These higher average credit scores occur *despite* these consumers having a shorter credit history — the length of credit history is an important input into credit scores, accounting for approximately 15% of FICO and 20% of VantageScore models (VantageScore, 2019, FICO, 2025). Because immigration status is not an input into credit scores, immigrants' higher scores reflect differences in credit quality between immigrants and non-immigrants.

3.3.3 Distribution of Credit Scores

The mean credit score may mask important heterogeneity in credit scores. Once scored, immigrants have a higher quality distribution of credit scores than non-immigrants of the same age. Panel B of Table 3 estimates the likelihood of having prime or higher credit scores, specifically, VantageScore > 660, where consumers with below prime or no scores are assigned a value of zero. Column 1 shows that immigrants are 4.6 percentage points more likely to have prime or higher credit scores by age 30 than their non-immigrant counterparts, increasing to 11.2 percentage points by age 40 (column 3) — a large increase relative to the baseline of 49.3% of non-immigrant consumers with prime or higher credit scores by age 40. There is heterogeneity by SSN Age, with immigrants who arrive later in life being half a percentage point less likely to have a prime or higher credit score by age 30 (column 2), primarily reflecting that later in life immigrants are less likely to have been scored by age 30. By age 40, effectively all consumers are scored, and later in life immigrants are approximately one percentage point more likely to have a prime or higher score per year of SSN Age (column 4). Appendix Table A3 shows that these results are similar when we use specifications that also include additional geographic fixed effects for *last* ZIP5, longest-held ZIP5, and the number of unique ZIP5s.

Panel C of Figure 2 shows how the fraction of consumers with prime or higher credit scores evolves over the lifecycle for each SSN Age cohort. In Panel C, "unscored" consumers—those without any credit score—are counted in the denominator, while Panel D conditions the plot on having a score. In Panel C, we observe two competing forces at play: On one hand, immigration cohorts arriving later in life are less likely to be scored, but on the other hand, they are more likely to have prime or higher credit once they are scored. In a cohort's twenties, the share of unscored consumers dominates the higher credit scores of the cohort's scored consumers, whereas by the cohort's thirties the higher credit scores of scored consumers dominate. Panel D shows that, conditional on having a credit score, immigrants are more likely to have prime or higher credit scores and the magnitude of these differences is relatively large.

Figure 1 displays the 10^{th}, 25^{th}, 50^{th}, 75^{th}, and 90^{th} quantiles of credit scores by age 40, calculated separately for immigrants and non-immigrants. This shows that immigrants have higher credit scores throughout the distribution. Consistent with this, Appendix Figures A8 and A9 show that immigrants have a higher quality distribution of credit scores than non-immigrants at different ages. The average immigrant is less likely to have a subprime (VantageScore < 600) credit score than the average non-immigrant at any observed stage in the life cycle. The average immigrant is more likely to have a very high credit score, as measured by prime plus (VantageScore > 720) or superprime (VantageScore > 780), by their mid-to-late thirties.[13] Appendix Figure A8 demonstrates that these findings hold irrespective of either the age of immigration or conditioning on having any credit score. Panels A and B of Appendix Figure A9 show that our findings hold for a consumer's first observed credit score, despite such scores often being based on limited credit data.

Overall, the results of this subsection indicate that the credit scoring system classifies immigrants, based on their credit behaviors, as higher credit quality on average than non-immigrants of the same age. There is also heterogeneity, with the immigrants who arrive later in life being higher credit quality than immigrants that arrive earlier.

3.3.4 Credit Delinquency & Credit Utilization

We also examine two alternative measures of credit quality that are common inputs to credit scores (Gibbs et al., 2025): any delinquency 90 or more days past due (conditional on holding any credit product), and the average credit card utilization rate (the ratio of the sum of credit card balances across cards to the sum of credit card limits across cards, conditional on holding any credit card).

The lifecycle of delinquency rates shown in Panel E of Figure 2 is consistent with our credit score results. SSN Age 18 consumers have higher delinquency rates than immigrants at all ages in our sample frame: from 21 to 40. Among immigrants, those arriving later in life generally have lower delinquency rates.

A complementary way to assess immigrants' creditworthiness is the "calibration bias" approach used in Bakker et al. (2025). Panel A of Figure 3 shows the delinquency rates for immigrants and non-immigrants by age 37, conditional on credit scores at age 30 for the consumers who have a credit score by age 30. Across all credit score groups, immigrants have lower average delinquency rates than non-immigrants. Panel A of Appendix Table A4 shows estimates from our regression

[13] Earlier in their life cycle immigrants are less likely to have very high superprime credit scores. This is because consumers typically have little information on their credit report when first scored — a "thin file" — and so the credit score for an entrant to the credit system typically takes a relatively small number of values, as shown by the CDF in Panel A of Appendix Figure A9. The longer a consumer's credit history, the more time there is for positive and negative informational events to occur, which shift a consumer's score up or down. More informational events allow for a more accurate measure of risk, leading to greater dispersion in credit scores.

specifications, where the outcome is any delinquency by age 37 and regressions include fixed effects for birth year, credit score at age 30, and ZIP5 at age 30. Column 2 of this table shows that immigrants' delinquency rates are 1.5 percentage points lower than for non-immigrants, or 5.5% of the non-immigrant mean (27.2%), and columns 3 to 12 show that the differences are most pronounced among consumers with lower credit scores at age 30. Thus, even conditional on credit scores at age 30, immigrants are less likely to become delinquent, suggesting that credit scores for immigrants may under-estimate their creditworthiness relative to non-immigrants.

Beyond the lower delinquency rates, an important reason credit scores are higher for immigrants is because they have lower credit card utilization. We illustrate this pattern in Panel F of Figure 2. Credit card utilization is consistently lower for immigrants, especially for later in life SSN Age cohorts. Both delinquency rates and credit card utilization suggest that immigrants are more creditworthy relative to the SSN Age 18 benchmark.

3.4 Access to Mortgages and Auto Loans

The previous subsections establish that once an immigrant has entered the credit system, they have higher average credit scores, lower delinquency rates, and lower credit card utilization than non-immigrants. These findings imply that, a few years after immigrating, the average immigrant is observably higher credit quality, with a higher credit score despite a shorter credit history than a non-immigrant who is born in the same year and enters in the same geographic location.

We now consider the timing of first credit access to mortgages and auto loans. We observe whether consumers access these products, and later discuss whether these equilibrium outcomes are driven by demand or supply factors. The majority of houses and vehicles in the US are purchased on credit and therefore are observed in credit reporting data (Gibbs et al., 2025).

Despite greater creditworthiness, immigrant credit access is lower than that of non-immigrants. We show the lifecycle of credit access for each type of credit in Figure 4 by SSN Age cohort for each age. Immigrants are less likely than non-immigrants to access credit at any point in the lifecycle. Panel C shows that immigrants are less likely than the SSN Age 18 non-immigrant benchmark to have had any credit card, although this difference largely disappears over the life cycle. We find that immigrants are *persistently* less likely to ever had access auto loan credit (Panel A) and mortgages (Panel B), even up to age 37, the end of the panel. Moreover, for auto loans and mortgages, there is a larger difference, relative to the SSN Age 18 non-immigrant benchmark, for immigrants arriving later in their twenties than for immigrants arriving earlier in life — the lines do not converge. This marked pattern suggests that delayed entry into the credit system leads to long-lived effects on credit access that go beyond observable measures of creditworthiness. That is, even though later in life immigrants have significantly higher average credit scores, their access to mortgages and auto loans lags significantly behind non-immigrants and even behind immigrants with an earlier immigration date.

We quantify this idea using our regression specifications, where the outcome is an indicator for the age of first credit access, separately for each type of credit as presented in Panel A of Table 4. We find that immigrants first access credit significantly later than non-immigrants: 5.0 years later for auto loans (column 1) and 2.3 years later for mortgages (column 3), on average. These average differences grow for later in life immigrants. One additional year of SSN Age predicts an additional delay of 0.6 years for a consumer's first auto loan (column 2), and 0.4 years for a consumer's first mortgage (column 4). Appendix Table A2 shows that these estimated delays in first accessing credit are robust to controlling for the consumer's last ZIP5, longest ZIP5, and number of ZIP5s.

To quantify the extensive margin – whether consumers access credit at all within our sample frame, we now change the outcome to whether a consumer has accessed a specific credit product (credit card, auto loan, or mortgage) by age of 37 (i.e., 8 years after the last SSN Age cohort arrives). Panel B of Table 4 presents the results showing large and significant differences in average long-run access to auto loans and mortgages: immigrants are 15.7 percentage points (20.6% of the outcome mean) less likely to have ever had an auto loan (column 1) and 8.9 percentage points (20.0%) less likely to have ever had a mortgage (column 3) by age 37. Columns 2 and 3 allow for heterogeneity in SSN Age and reveal how the lower use of auto loans and mortgages for immigrants, compared to non-immigrants, is substantially larger for the immigrants who arrive later in life. These results are robust to including additional fixed effects (Appendix Table A2), including for credit score at age 30 as shown in Panels B and C of Figure 3 (for regression estimates, see Panel B of Appendix Table A4).

The difference in mortgage access between immigrants and non-immigrants in Panels A and B is largely explained by the timing of immigrants' arrival in the United States, with lower access emerging for consumers with later in life SSN Ages (see Appendix Figure A6 for heterogeneity by each SSN Age) and controlling for the last observed geographic location (Appendix Table A2). A large immigrant versus non-immigrant difference remains in auto loan access, which we explore in Section 5 by studying ACS and SCF data.

3.5 Access to Credit Cards

We now examine consumers' access to credit cards, a key source of unsecured borrowing in the United States. We use our regression specifications to study the extensive margin of credit card access, with outcomes being an indicator for the age of first accessing a credit card, and whether a consumer has ever held a credit card by age 37. Estimates are reported in Panel A of Table 4, where we find a significant immigrant to non-immigrant difference in the average age of accessing credit cards: 4.8 years later for immigrants first taking out a credit card (column 5), and the difference is greater for later in life immigrants (column 6). By age 37, these differences in credit card access have largely disappeared: Panel B of Table 4 shows that immigrants are only slightly, 1.4 percentage points, less likely to have a credit card than non-immigrants (column 5) with an economically small but significant difference by SSN Age (column 6). The patterns in Panel C of Figure 4 also suggest that credit card access of immigrants catches up more quickly than access to auto loans and mortgages (Panels A and B).

We now look at the intensive margin of access to credit cards, examining how the number of credit cards and their limits evolve over the consumer lifecycle for different immigration cohorts. Panel D ofFigure 4 shows the number of credit cards, Panel E shows the total value of credit limits (with additional conditional measures in Panels C and D Appendix A7), and Panel F shows the credit card limit per card conditional on holding any card. Consistent with the evidence on access to auto loans and mortgages, credit card limits of immigrants with later in life SSN Ages take many years to catch up to immigrants with earlier in life SSN Ages and non-immigrants. For example, in Panel E of Figure 4, the SSN Age 29 cohort takes until age 35 to converge to the same total credit limit as SSN Age 18 consumers. This long delay in convergence reflects different dynamics for the number of credit cards (Panel D) and the average credit limit per card (Panel F). In Panel F, the average credit limit per card *never* catches up to earlier cohorts (i.e., by age 40, the SSN Age 29 cohort has the lowest mean credit limit per card of all immigrant cohorts). By contrast, the number of credit cards catches up and surpasses SSN Age 18 cohort within five years of immigration. Panel B of Appendix Table A3, presents the corresponding estimates from our baseline cross-sectional regressions at age 30 (when some cohorts have not yet crossed the non-immigrant line) and at age 40 (when all immigrant cohorts have crossed). Our credit card results are robust to controlling for credit score at age 30 as shown in Panels D to F of Figure 3 (for regression estimates, see Panel B of Appendix Table A4).

4. Paired Cohorts

Immigrants and non-immigrants differ along many dimensions. To account for this, we develop and apply a *paired cohort* strategy, which focuses comparisons on two immigrant cohorts, who are the same age but arrive in the United States a single year apart. This estimation approach is not a causal design, but allows us to quantify the dynamics of how immigrating one year later in life correlates with credit outcomes.

4.1 Methodology

While we have so far defined a cohort as all consumers with the same SSN age, in this section we define a cohort more narrowly, adding the requirement that all consumers with the same SSN Age are also the same age. Thus, we define a cohort c, as the set of consumers who have the same birth year *and* have the same year of SSN assignment—e.g., the set of people born in 1985 and who also immigrated in 2007 and were therefore assigned an SSN at age 22 is a cohort in this analysis (birth year 1985 $\times$ SSN Age 22). We aggregate the data such that there is one observation per cohort and year from 2000 to 2024. We then restrict to cohorts where $2003 \leq Birth\,Year + SSN\ Age \leq 2011$ to ensure that we observe pre-periods before a cohort enters the BTCCP.

A pair p, is a set of two cohorts $\{k', k''\}$, consisting of a focal cohort k', and a matched control cohort k''. Cohorts within a pair have the same birth year, but the SSN Age for the control cohort k'' is one year earlier in life: $SSN\ Age_{k''} = SSN\ Age_{k'} - 1$. For example, our focal Birth Year 1985 $\times$ SSN Age 22 cohort is matched to a control of Birth Year 1985 $\times$ SSN Age 21 cohort. To implement this strategy, we restrict attention to focal cohorts of $SSN\ Age$ 22+ to ensure the availability of an immigrant cohort as a control. There are 68 pairs of cohorts containing consumers who had their SSNs assigned between 2003 and 2011, and we trim the data to ensure that our panel is balanced within and across pairs. We define event time t based on the focal cohort, keeping 16 years of annual data from two years prior to the year when the focal cohort's Age = SSN Age, to thirteen years after. Therefore, our dataset consists of 2,176 cohort-by-year observations with a paired structure. In our regressions we weight each observation by the size of its cohort.

We estimate the following specification via OLS:

$$\mathrm{Y}_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_\tau \big(\mathbb{I}\{focal\ cohort\}_{k(p)} \times \mathbb{I}\{t = \tau\}\big) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \tag{3}$$

where $Y_{k(p),t}$ is the outcome variable in event time t for cohort k belonging to pair p. This specification is akin to difference-in-differences where the indicator $\mathbb{I}\{focal\ cohort\}_{k(p)}$ plays the role of the treatment variable, and the interacted event time indicators $\mathbb{I}\{t = \tau\}$ implement a dynamic difference-in-differences, relative to the omitted event time period $\tau = -2$. Given this, δ_τ shows how credit outcomes, observed in the same year within a pair, develop for each cohort relative to consumers with the same birth year who immigrated one year earlier. To focus on variation within a pair and to account for pair-specific trends, the specification includes fixed effects for each cohort in a pair $\pi_{k(p)}$ and pair-by-event time fixed effects $\nu_{p,t}$.

Although event time t is the focal cohort's year of SSN assignment, the focal cohort is "treated" with one year less in the U.S. than the control cohort. To emphasize this non-standard timing in the plots, we label event time $t-1$ as "C" and event time t as "T" to respectively denote the years of SSN assignment for the control and focal cohorts.

We calculate 95% confidence intervals using wild cluster bootstrapping (using Webb weights and 10,000 iterations) by birth year to allow for dependence over the lifecycle and for dependence within pairs. Such clustering is especially important in our context because one cohort can be a focal cohort in one pair but also act as a control group in another pair. We use the wild cluster approach because we only have 13 clusters.

4.2 Results

We observe distinct dynamics for credit market entry versus later-in-lifecycle credit access using the paired cohort approach. Figure 5 presents the results on accessing any credit product or credit card (Panel A), access to auto loans and mortgages (Panel B), and credit card limits (Panel C). Reinforcing the results in the preceding sections, we observe a temporary difference in the relationship between one year later in life immigration and accessing credit cards. In the year the focal cohort receives its SSN — event time T — focal immigrants are 20 percentage points less likely to have any credit card relative to control immigrants (same-age immigrants with an SSN Age one year lower). This difference disappears by year four and results are near-identical for accessing any credit product.

In contrast, access to mortgage and auto loans does not converge. Thirteen years after immigrating, immigrants, relative to same-age immigrants who were assigned an SSN one year earlier in life, are 1.11 percentage points (95% C.I. −1.31 to −0.93) *less* likely to have had an auto loan and 1.36 percentage points (95% C.I. −1.78 to −0.93). Thus, even a one-year difference in SSN Age gives rise to the long-run differences that we documented in our earlier cross-sectional analysis.

Table 5 displays the cumulative years of delay in accessing credit by event times C, T, $T+5$ and $T+10$, with Appendix Figure A10 showing the full estimates.[14] We estimate that one year later in life SSN Age is associated with cumulatively 0.73 to 0.78 years less credit history relative to same-age immigrants, and most of this delay is realized by the focal cohort's event time $T+1$. This finding is consistent with the idea that later in life immigration cohorts' initially delayed credit access results in persistently shorter credit histories.

The results for auto loans and mortgages differ in terms of magnitude and dynamics. Thirteen years after the focal cohort's immigration, we estimate an average delay of 0.28 years for first mortgages, and a delay of 0.44 years for first auto loans from one year later in life immigration relative to same-age immigrants. The shorter average delay for credit products typically accessed later in life suggests that immigrants manage some degree of catch up over time. However, this catch up is not complete as of the end of the event window, and as the results in the previous section suggest, some immigrants might not access auto loans or mortgages at any point in their lifecycle, potentially related to their later in life immigration.

Turning to the initial dynamics, only a small fraction of the delay in auto loan or mortgage access can be attributed to "mechanical delay". Instead, most of the delay emerges several years after their year of SSN assignment, lining up with when in the lifecycle these products are typically accessed for the first time by non-immigrants. When taken together with the fact that immigrants in their thirties who immigrate at a later stage in their twenties have *higher* average credit scores throughout this period (Figure 2), the later-emergence and persistence of the delay in auto loan and mortgage access constitute consequential longer-term outcomes correlating with immigrating a single year later in life.

Finally, we see a non-monotonic difference in credit card limits for paired immigration cohorts, as displayed in Panel C of Figure 5, and a similar pattern is observed in the number of credit cards in Panel D of Appendix Figure A10. For the first eight years after the cohorts immigrate, we observe that the focal cohort has significantly lower credit card limits. This difference is at its widest two years after the focal cohort's year of SSN assignment, with average credit limits being \$1,705 (95% C.I. −\$1,936 to −\$1,478) *lower* than those of the control cohort. The difference

[14] This is calculated by accumulating coefficient estimates over the event window. For example, if we obtained an estimate of -0.20 in event time T, -0.05 in event time $T+1$, and -0.03 in event time $T+2$ for credit card access, we could say that the focal cohort is delayed by an average of 0.20 years as of event time T, 0.25 years by event time $T+1$, and 0.28 years by event time $T+2$.

shrinks over time to become indistinguishable from zero a decade after the focal cohort's year of SSN assignment. There is a similar pattern for the number of credit cards, crossing a few years earlier.[15] This nonlinearity reflects the overarching tension that we find throughout this paper. On one hand, we find consistent evidence that immigrants and especially those who immigrate later in their twenties have substantially higher creditworthiness. Yet, their later entry into the U.S. credit system leaves them with less credit history. Thus, despite their better credit quality, their access to credit lags behind, in the case of auto loans and mortgages for at least thirteen years after immigrating.

5. Mechanisms

We now discuss potential mechanisms that could lead to reduced credit access for immigrants, despite their higher measured creditworthiness. The subsections that follow respectively consider creditworthiness, emigration, credit demand and tastes, and history dependence as potential mechanisms.

5.1 Creditworthiness

We have shown that immigrants (and especially later-in-life immigrants) typically have better credit scores and exhibit lower risk profiles on other credit behaviors (lower delinquency rates and lower credit utilization), which indicates that **creditworthiness** cannot explain our results. Higher credit scores do not lead to lower long-term access to auto loans and mortgages, nor would they delay higher credit card limits by a decade. Further evidence against creditworthiness driving reduced credit access is provided by a calibration bias approach, in which we consider credit outcomes at age 37 while controlling for credit scores at age 30. If measured creditworthiness explained subsequent credit access, then the coefficients on immigrant status and age of immigration should attenuate to zero conditional on credit scores. However, Panel B of Appendix Table A4 shows that the reduced credit access for immigrants shown in Table 4 is still present after adding this credit score control. Immigrants have higher credit scores despite lower initial credit limits and being persistently less likely to hold auto loans and mortgages, factors we might expect to make them less risky.

Moreover, immigrants are systematically less likely to be delinquent by 37 in all credit bins

[15] Two years after the year of SSN assignment, the focal cohort has 0.64 (95% C.I. −0.68 to −0.59) *fewer* credit cards, on average, than the control cohort's average. This negative estimate has fully dissipated after ten years. After thirteen years, the focal cohort's average is 0.06 (95% C.I. 0.05 to 0.07) *higher* than the control cohort's average.

below Prime Plus, even after controlling for credit score at 30 (as shown in Panel A of Appendix Table A4). As a final piece of evidence against creditworthiness driving the reduced credit access of immigrants, Panels B and C of Figure 3 (with estimates in Appendix Table A4) show lower access to auto and mortgage credit by age 37 for immigrants across every credit score group.

5.2 Emigration

A different possible explanation for some of our results, especially lower long-term access to mortgages, could be **emigration**: consumers expecting to emigrate may be less likely to purchase property. Even if immigrants and non-immigrants were similarly likely to purchase property each year, emigration would mean that immigrants are present in our data for fewer years, making catch-up less likely.

Our findings remain robust after accounting for this differential attrition of immigrants. In Appendix Figure A11 and Appendix Table A5 we condition on consumers still present in the data in 2024 and examine all of the main credit outcomes; the results are consistent with those in our main Figure 4. Therefore, while emigration may be a factor that influence some credit decisions, it cannot drive the differences in credit access that we document.

A final related concern is that the *possibility* of future emigration leads consumers to avoid investing in assets that require financing. However, the relative likelihood of emigration across cohorts is inconsistent with this concern: later-in-life immigrants are *less* likely to emigrate than earlier-arriving immigrants (see Appendix Table A5), but have *lower* mortgage and auto credit access in the paired cohort analysis.

5.3 Credit Demand and Tastes

Another possible mechanism driving lower immigrant access to credit is that they have lower **credit demand** than non-immigrants. In Panel E of Appendix Figure A7 we examine the outcome *months since last inquiry*, where fewer months since last inquiry indicate higher credit demand. On average, immigrants in their twenties have substantially higher credit demand than non-immigrants. However, during their thirties, this pattern reverses and immigrants have slightly lower demand. Panel D of Appendix Figure A10 displays a similar pattern using the paired cohort design: the focal cohort initially has clearly higher credit demand, and it is only in year seven that they begin to have slightly lower demand. Moreover, the lower credit demand we observe later in the lifecycle may result from being deterred by earlier rejections. Thus, this evidence suggests that delayed access to credit is not purely an artifact of delayed demand for credit for later-in-life SSN Age cohorts. A limitation of our data is that we cannot observe the type of credit product applied for.

Differences in asset-linked credit – auto loans and mortgages – could also reflect **different tastes** for the underlying asset. That is, immigrants could value cars less than non-immigrants or be more likely to prefer renting over owning. To examine this, we use repeated cross-sections from the American Community Survey (ACS), which allow us to observe asset ownership directly — cars and homes — and to control flexibly for detailed demographics, including income.

Figure 6 plots estimates from regressions of mortgage status (Panel A), homeownership (Panel B) and car ownership (Panel C) on immigrant status across four waves of the ACS spanning 2006 to 2023. During this period, immigrants arriving in their twenties are significantly and persistently less likely to have a mortgage. Moreover, these differences exist *after* controlling for birth year, geography, education, race, Hispanic ethnicity, and survey year. Controlling for log household income, and adding further controls for employment and occupation, only slightly attenuates estimates, indicating that the observed difference is not income-driven. However, immigrants' reduced mortgage access declines as the cohorts in our sample accumulate more time in America. Patterns in homeownership are similar, a result consistent with Borjas, 2002. This finding is consistent with mortgage credit often being necessary to become a homeowner and rules out that the mortgage gaps reflect immigrants being wealthier and purchasing property outright with cash. We observe initial gaps in auto ownership, but these gaps do not persist in the long run, and the ACS does not contain a variable for auto loan use.

Differences in asset-linked credit — auto loans and mortgages — could also reflect that immigrants have *different tastes for financing* cars or homes. We investigate this using the 2022 Survey of Consumer Finances (SCF), the only vintage that identifies immigrants.[16] Like the ACS, the SCF allows us to control for income and other demographics, but it also asks about attitudes toward debt. Columns 1 and 2 of Table 7 show that immigrants report a much more negative attitude towards auto debt than non-immigrants. Specifically, immigrants are 15 percentage points less likely than non-immigrants to agree that it is "all right for someone like yourself to borrow money" to fund an auto purchase, and this variable strongly predicts auto loan use (Columns 5 and 6). Unfortunately, there is no analogous SCF question about attitudes to credit card or mortgage debt. However, there is a question about general attitudes to debt, and the estimates reported in columns 7 and 8 show no clear evidence of a more general immigrant "debt aversion" (e.g.,

[16]See Appendix A.2. for details about the SCF.

Martínez-Marquina and Shi, 2024).[17]

5.4 History dependence

A plausible mechanism for our findings is that the length of a consumer's credit history itself may be an important factor in lending decisions, as immigrants arrive with no credit history and cannot be scored, while later-in-life immigration necessarily reduces the length of credit history. This mechanism is consistent with our evidence of lower immigrant credit access despite higher creditworthiness than *same-age* non-immigrants. **History dependence** of credit is consistent with the differences in credit access observed in the paired cohort analysis.[18]

However, we cannot be certain that the key mechanism for delayed immigrant credit access is only the number of years in the United States credit system, because there are also other sources of history dependence. Our results are organized around life cycles defined by consumers' birth years, however, it is possible that immigrants' life cycles are instead more closely linked to when they arrive in the United States — when their American labor market and immigrant experience begins — which is a different type of history dependence. Being in the country for an additional year means that the paired control cohort has an additional year to potentially take out credit, an extra year of U.S. earnings, and is a year further along their personal and professional American life cycle, which, in turn, may mean they are a year further along in the credit demand life cycle. Such differences may be most pronounced in the initial years after arrival, but if the differences have longer-term persistence, they might contribute to the absence of full catch-up in some credit access outcomes. History dependence is also consistent with the finding that the difference at age 37 in ever having had a mortgage, a product often not accessed by consumers until their thirties, is driven by differences in immigrant age of SSN assignment in their mid twenties — a delay in starting the U.S. lifecycle translates into a delay a decade later.

To examine history dependence in earning ability, we again turn to the ACS. Table 6 shows that, after controlling for demographics, immigrants arriving in their twenties have substantially lower household income (17% in 2006 to 2013 and 8% in 2019 to 2023) than non-immigrants

[17] Appendix Table A8 contains the other available measures of attitudes to debt in the SCF, with no clear evidence of a more general aversion to debt by immigrants. Panel A of Appendix Table A8 examines SCF questions about whether respondents have applied for credit over the preceding twelve months, a measure of credit demand, however, we do not have the statistical power to detect differences between immigrants and non-immigrants.

[18] If the length of credit history itself is the mechanism (separately from its effects on credit scores), then retaining positive credit information for shorter periods may be a more efficient credit market design, as indicated by the Kovbasyuk and Spagnolo (2024) model. Blattner et al. (2022) find that shorter credit histories have trade-offs, reducing credit for some existing borrowers but helping new consumers to enter the credit market.

and that this gap is larger for immigrants who immigrate later in their lifecycle.[19] These income differences may contribute to lower access to asset-linked credit, either through demand or supply. Figure 6 and Appendix Table A7 show that even after controlling for income and other factors, persistent differences remain. Each additional year of later-in-life immigration leads to persistently lower mortgage use, homeownership, and, to a lesser extent, car ownership. This persistence after controlling for income points toward additional frictions beyond earnings capacity.

To the extent that having no or a limited credit history is a financial friction for immigrants that rations credit, market solutions using alternative data in lending decisions are emerging to expand credit access. American Express has linked their internal data internationally, allowing their foreign customers to easily apply for a U.S. American Express card. FinTech lenders, such as MPower Financing and Prodigy Finance, provide finance for the cost of educating foreign students at select American universities (Bloomberg, 2025). More generally, FinTech firms (e.g., Nova Credit) and credit reporting agencies have recently developed technological solutions to enable access to foreign credit reports and to translate foreign credit scores into consistent formats that lenders can use (FinTech Magazine, 2024, TransUnion, 2018, Equifax, 2024).

6. Conclusion

This study offers the first large-scale empirical examination of immigrants' assimilation into the U.S. consumer credit system. We document how immigrants compare to non-immigrants and to immigrants arriving later in life in terms of their creditworthiness and credit usage as they age. Upon credit market entry, immigrants are strongly positively selected in terms of their creditworthiness. By age 30, immigrants' average credit scores are 26.7 points higher than non-immigrants, and this difference persists with age. Yet, despite immigrants' superior measured creditworthiness, we show that by age 37, immigrants are 15.7 percentage points less likely to have ever had an auto loan and 8.9 percentage points less likely to have ever had a mortgage, compared to non-immigrants of the same age in the same geographic area.

Our analyses also show persistent differences in credit access between same-age immigrant cohorts arriving at different times, even when arriving just a single year later. The pattern of our

[19] This pattern of lower income for immigrants is consistent across the distribution, as shown in Appendix Figure A1. These results are robust to additional fixed effects for employment status and occupation, presented in Appendix Table A6. Such results of persistently lower income based on the timing of arrival in America are broadly consistent with Chiswick (1978) and causal estimates in Kerr et al. (2026), and Lubotsky (2007) shows that estimates using repeated cross sectional data may under-estimate differences.

results is consistent with a role for history dependence in immigrant credit access. In light of the major contributions of immigrants in credit-dependent areas such as entrepreneurship, our results highlight an opportunity for market innovations to expand lending to immigrants in America.

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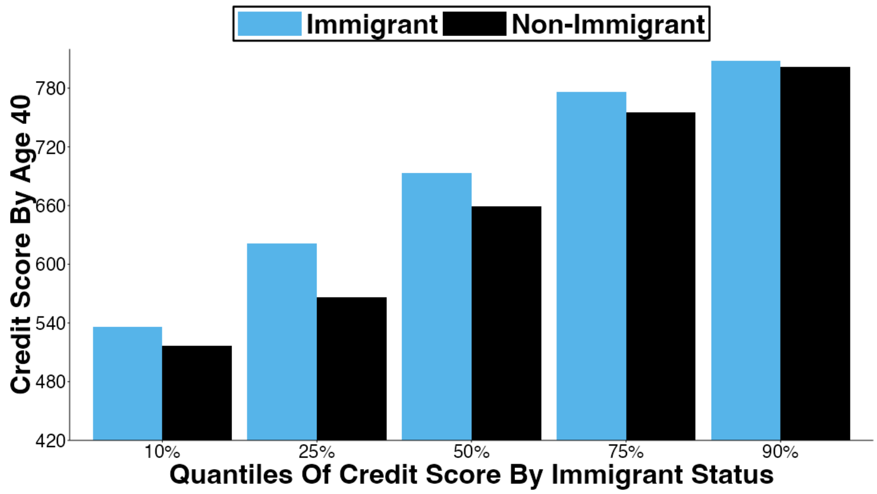
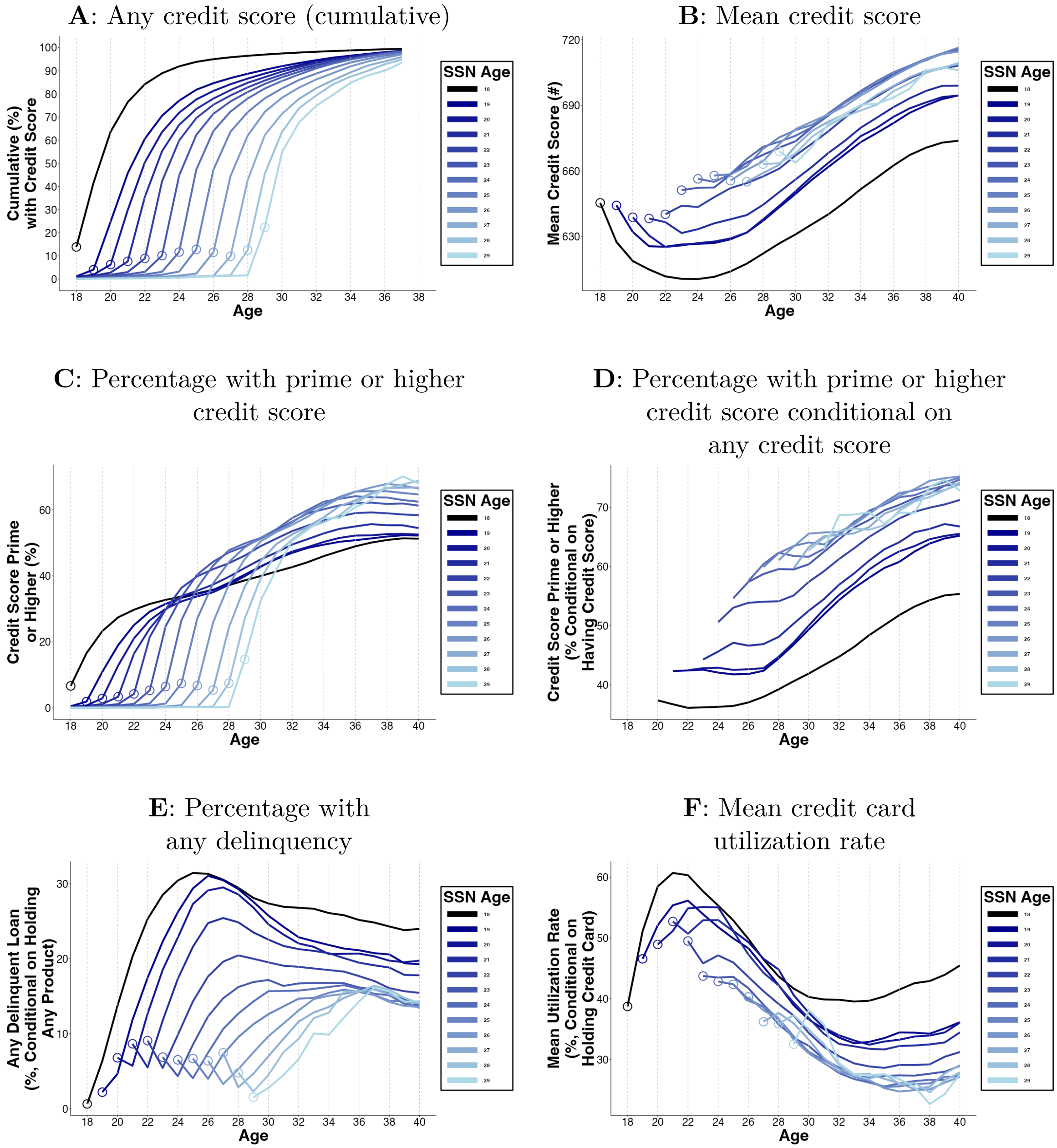
Figure 1: Quantiles of credit scores by age 40 by immigration status



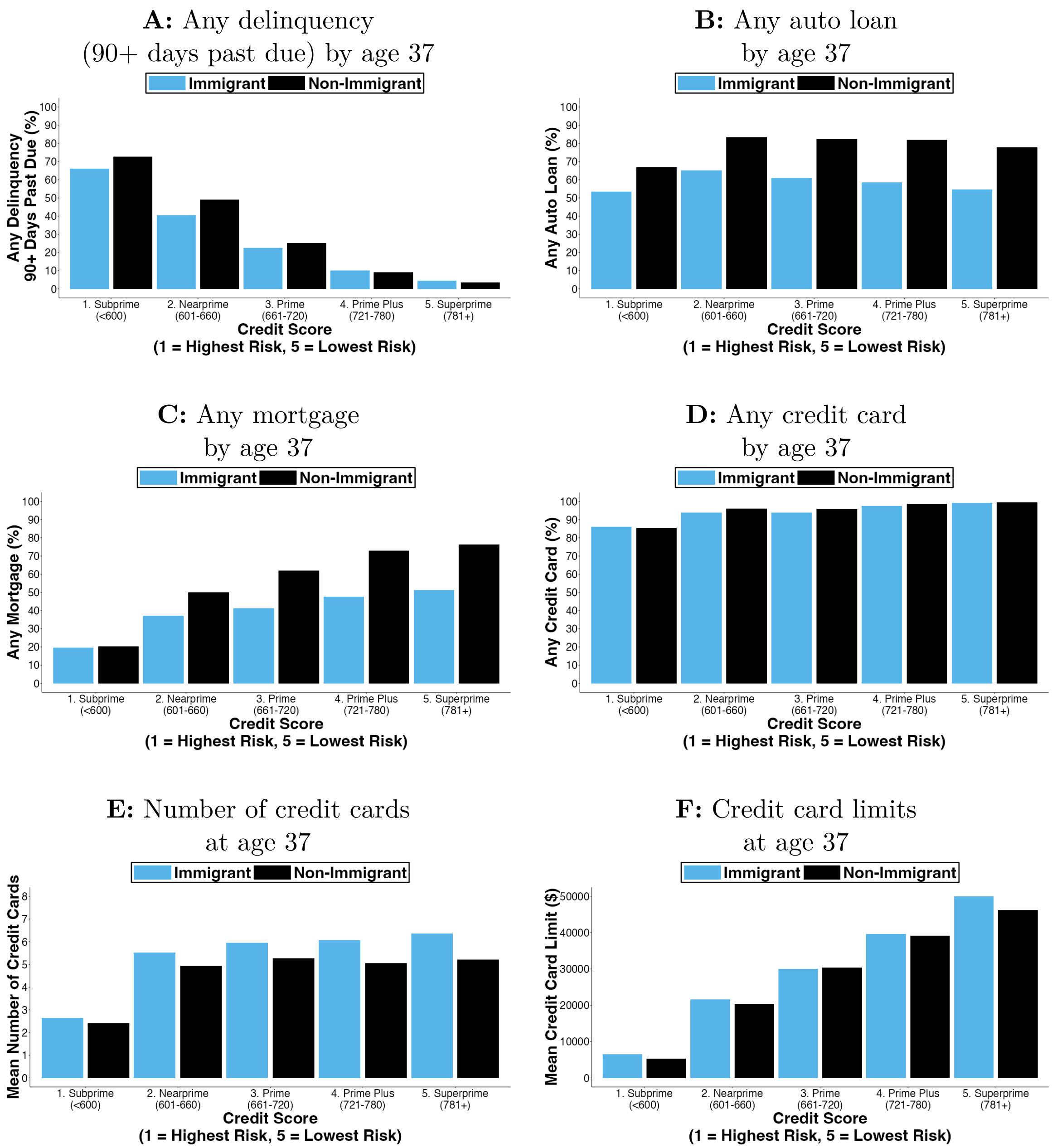
This figure shows the values of credit scores by age 40 taken at the 10%, 25%, 50%, 75%, and 90% quantiles of the distribution. These quantiles are calculated separately for immigrants and non-immigrants. We classify immigrants as those assigned SSNs at ages 21 to 29, and non-immigrants as those assigned SSNs at age 20 or younger. The data consists of 5,906,217 consumers with non-missing credit scores by age 40, of which 336,189 are immigrants and 5,570,028 are non-immigrants. For consumers who are not observed in the data at age 40, this uses their credit score at the oldest age before 40 that is observed.

Figure 2: Lifecycle of creditworthiness by SSN Age cohorts



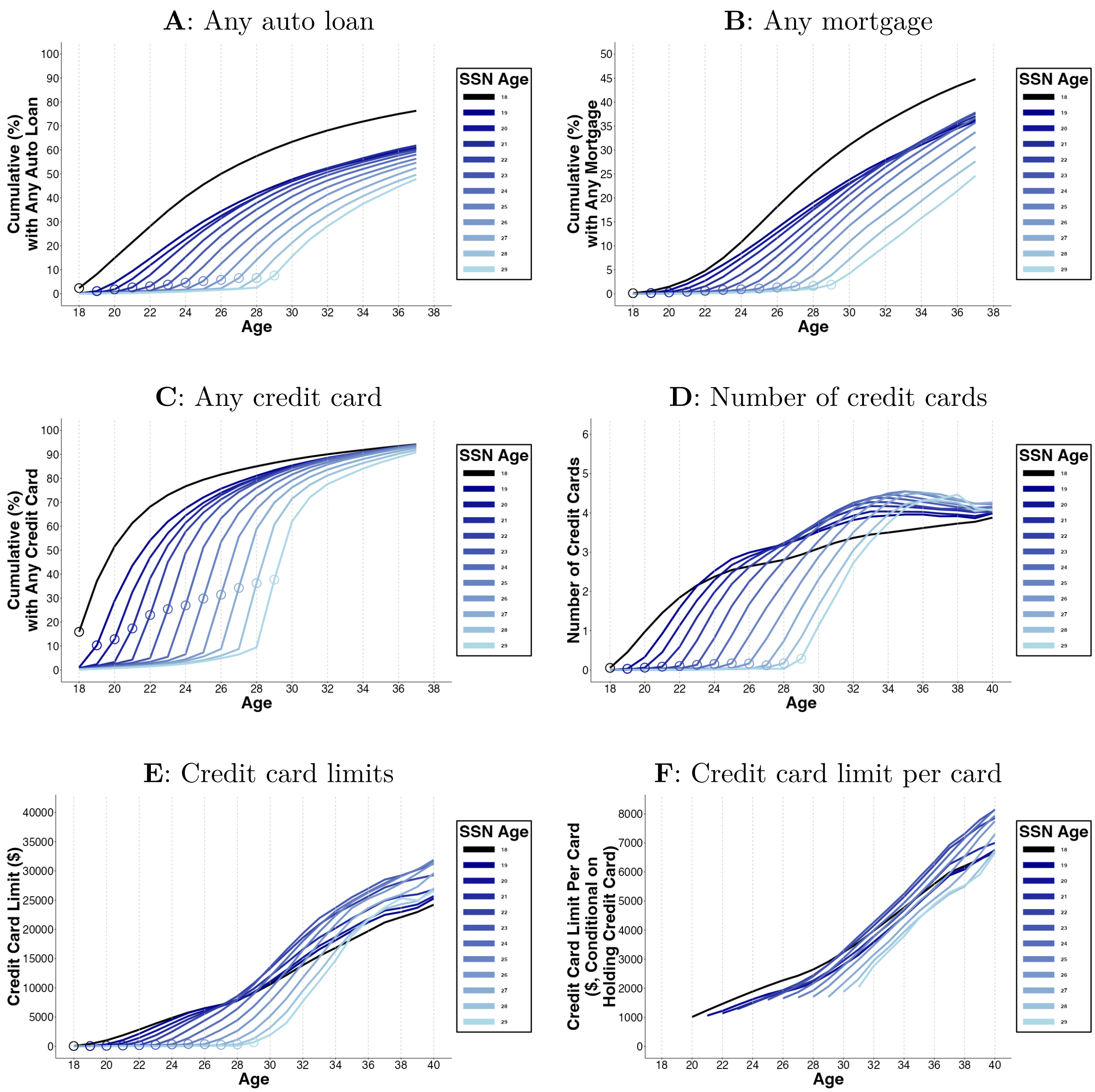
This figure plots outcomes at different calendar ages for each SSN Age cohort. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later in life SSN Age cohorts from 19 to 29. We classify immigrants as the SSN Age cohorts 21 to 29. $Age = SSNAge$ is indicated by the circles on each line. This figure uses data for birth years from 1982 to 1987. Because Panel A is a cumulative chart, we end it at age 37, whereas for the other panels the estimates for ages 38-40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for ages 40, 39 to 40, and 38 to 40, respectively, by the end of our data in 2024. Panel A presents the cumulative fraction of consumers that have been credit scored by each age. In panels A to D, credit scores are measured by VantageScore. In Panels C and D, Prime or Higher Credit Score is a VantageScore of 660 or higher. Panel E is conditional on holding any credit product; Panel F is conditional on holding a credit card.

Figure 3: Credit outcomes by age 37 conditional on credit score at age 30



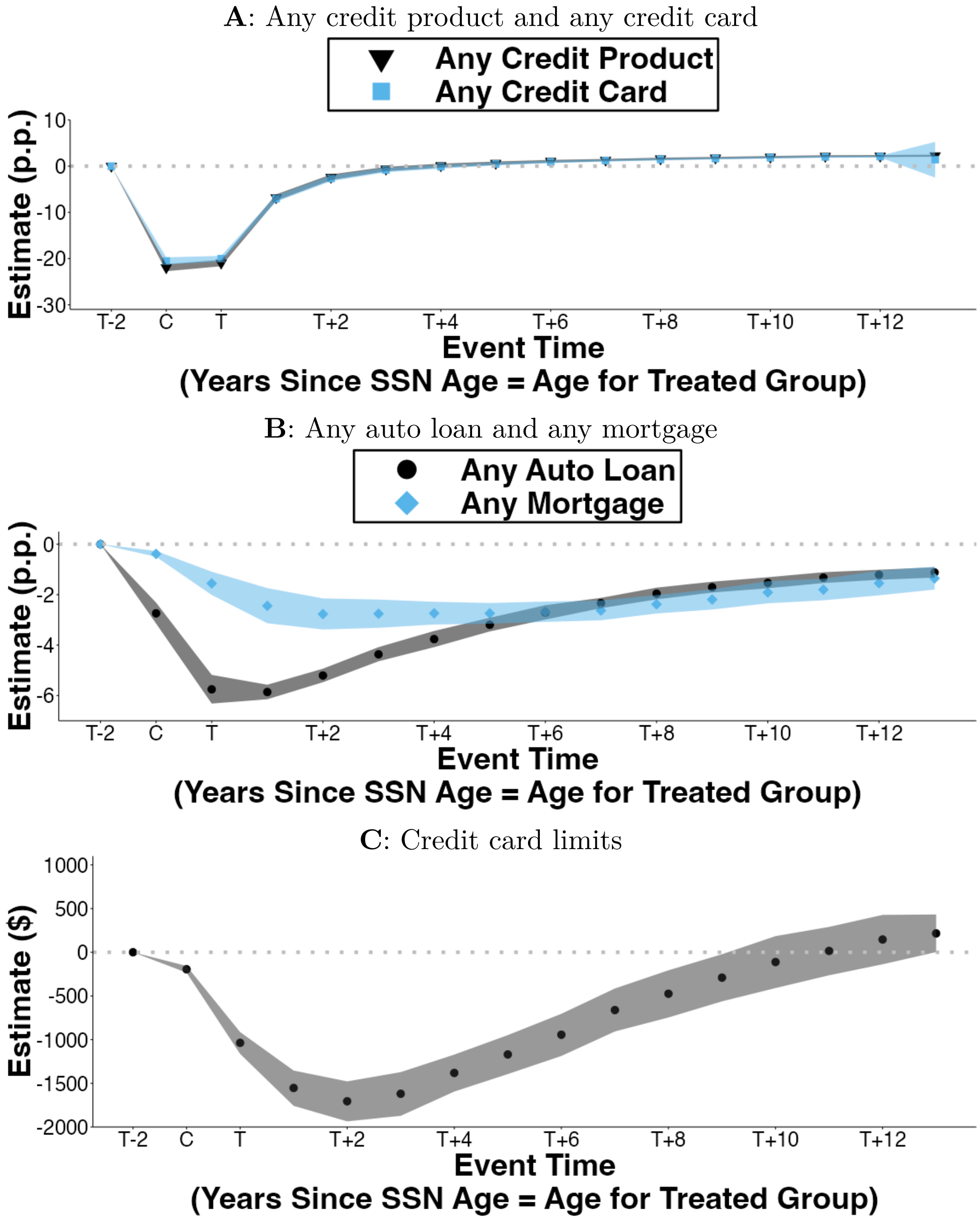
Each panel shows mean outcomes measured at age 37 conditional on a consumer's credit score measured at age 30. Each panel splits results by immigrants and non-immigrants, where immigrants are those with a SSN Age of 21+.

Figure 4: Lifecycle of credit access by SSN Age cohorts



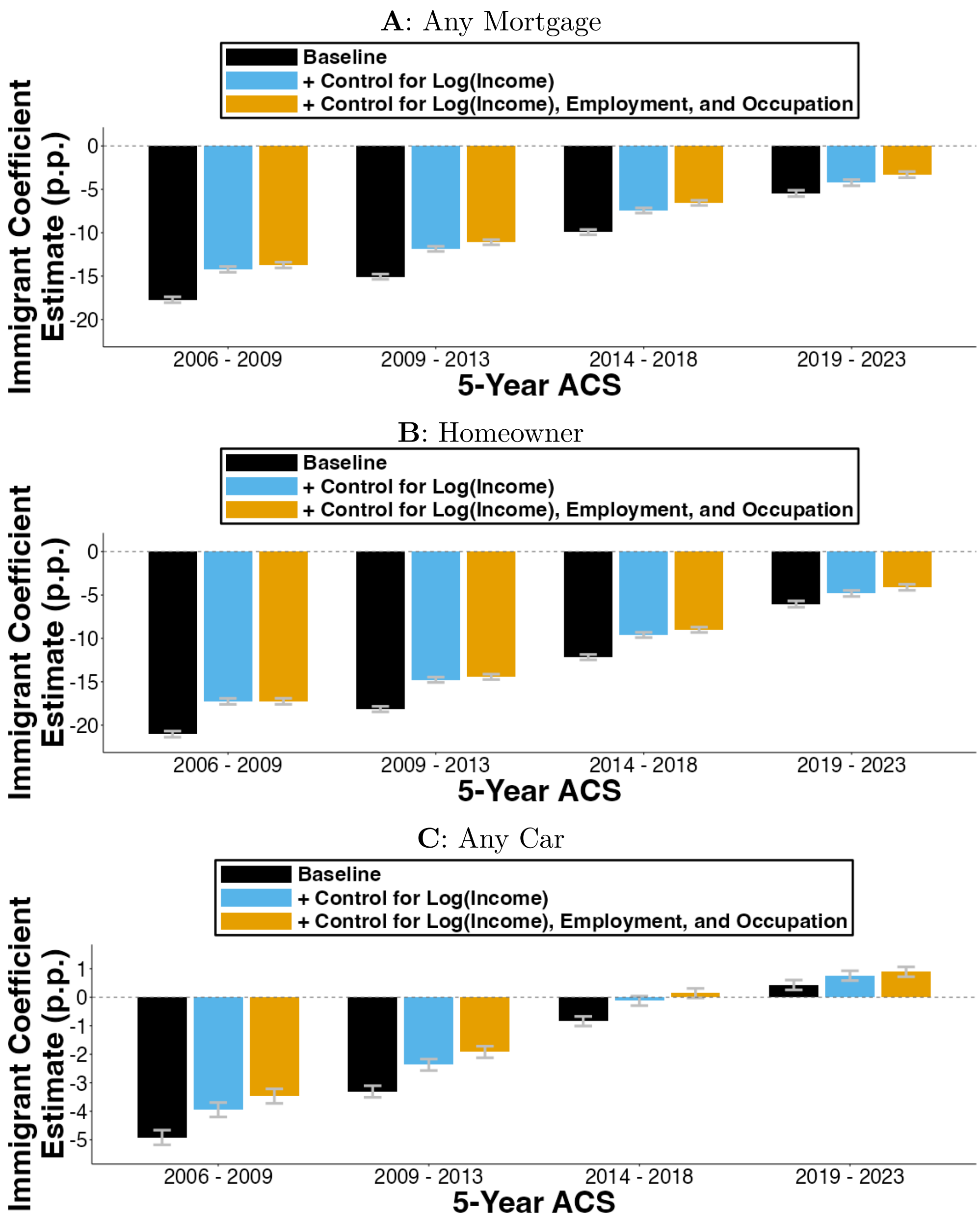
This figure plots credit outcomes at different calendar ages for each SSN Age cohort. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later in life SSN Age cohorts from 19 to 29. We classify immigrants as the SSN Age cohorts 21 to 29. $Age = SSNAge$ is indicated by the circles on each line. Panels A, B, and C use data for birth years from 1975 to 1987. Because Panels A, B, and C are cumulative charts, we end them at age 37, whereas for the other panels the estimates for ages 38-40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for 40, 39 to 40, and 38 to 40, respectively, by the end of our data in 2024. Panels D, E, and F use data for birth years from 1982 to 1987 and plot means of the outcomes.

Figure 5: Paired cohorts: Dynamics of credit access



This figure presents dynamic estimates for differences in credit access between a cohort with SSN Age $= s$ and a matched same-age control cohort with SSN Age $= s - 1$. The differences are presented in event time, where C is the year when age equals SSN Age for the $s - 1$ (control) cohort and event time T is the year when age equals SSN Age for the s (treatment) cohort. Panels A and B show estimates, in percentage points, from regressions where the binary outcomes measure any credit product (black) and any credit card (blue) in Panel A, and in Panel B, any auto loan (black) and any mortgage (blue). Panel C shows estimates for the outcome, the value credit card limits ($). The shaded areas indicate 95% confidence intervals, calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations.

Figure 6: American Community Survey: credit access



This figure uses public data from the 5-year American Community Survey (ACS) waves 2009, 2013, 2018, and 2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each bar shows estimates from a separate cross-sectional regression. Each bar shows coefficient estimates on an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Each x-axis category shows results from a different 5-year ACS wave. The outcome in Panel A is whether the consumer has a mortgage, in Panel B it is whether they are a homeowner, and in Panel C is whether they have an auto. All regressions include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, and survey year. The blue bars also include a control for Log (Income), the log of household income that is winsorized at 99.9th percentile and is scaled to mean zero and standard deviation one. The orange bars additionally include fixed effects for employment status and for occupation. Units are percentage points. 95% confidence intervals are calculated using heteroskedasticity-robust standard errors.

Table 1: Summary statistics on immigrants and non-immigrants

This table presents means and counts for the number of consumers (N) these are calculated from. Means are calculated separately for non-immigrants ($SSNAge < 21$) and immigrants ($SSNAge$ 21 to 29) in our data. *SSN Age* is the age that a consumer is assigned an SSN. The variables *Credit Score by Age 30* and *Credit Score by Age 40* and are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). The variables *Any Score by Age 30* and *Any Score by Age 40* are whether a consumer had any score by those ages (for consumers not observed at a given age, we use whether they had a score at the oldest age observed before that age). The variables *Any Auto Loan*, *Any Mortgage*, and *Any Credit Card* indicate whether, up to the end of our data in 2024, the consumer is ever observed to have any auto loan, mortgage, or credit card respectively. The variables *Age at First Auto Loan*, *Age at First Mortgage*, and *Age at First Credit Card* are each conditional on the consumer ever having auto loans, mortgages, credit cards respectively at any point up to the end of our data in 2024. The variable *Credit Card Limits at Age 40* has smaller sample sizes because birth years 1985, 1986, and 1987 are not observed at age 40.

	Non-Immigrants *(SSN Age < 21)*		Immigrants *(SSN Age 21 to 29)*	
	Mean	*N*	Mean	*N*
Credit Report				
Age at First Credit Report	20.19	5,914,891	26.09	384,396
Credit Score				
Any Score by Age 30 (%)	96.26	5,914,891	81.63	384,396
Credit Score by Age 30	624.2	5,693,584	658.3	313,799
Any Score by Age 40 (%)	99.35	5,914,891	97.94	384,396
Credit Score by Age 40	658.1	5,876,224	687.5	376,461
Auto Loan				
Any Auto Loan (%)	81.14	5,914,891	65.62	384,396
Age at First Auto Loan	26.13	4,799,057	30.59	252,243
Mortgage				
Any Mortgage (%)	50.28	5,914,891	44.42	384,396
Age at First Mortgage	29.53	2,973,846	32.88	170,754
Credit Card				
Any Credit Card (%)	96.87	5,914,891	97.39	384,396
Age at First Credit Card	22.33	5,730,022	27.20	374,365
Credit Card Limits at Age 30	\$11,492	5,914,891	\$10,211	384,396
Credit Card Limits at Age 40	\$22,112	4,579,356	\$26,608	342,258

Table 2: Timing of credit market entry

Columns 1 and 3 in this table are estimates from the cross-sectional OLS regression specified in Equation 1 that includes an indicator for immigration status (*Immigrant* equals 1 if the consumer's SSN was assigned at age 21 or older). Columns 2 and 4 in this table are estimates from the OLS regression specified in Equation 2 that contains both the indicator for immigration status, and also *SSN Age*, the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group). All regressions include fixed effects for the birth year of the consumer and fixed effects for consumers' first observed ZIP code (First Zip5). The outcome in columns 1 and 2 is age at first credit report, and the outcome in columns 3 and 4 is age at first credit product. In these regressions, consumers who never have a credit report or never have a credit product are assigned their age as of 2025. The units in this table are years of age. Standard errors are clustered by birth year × the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Dep Var: Age at First...	Credit Report		Credit Product	
	(1)	(2)	(3)	(4)
Immigrant	5.53***	2.29***	5.20***	2.07***
	(0.02)	(0.02)	(0.02)	(0.03)
SSN Age		0.75***		0.73***
		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.193	0.209	0.118	0.126
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	20.19	20.19	21.56	21.56

Table 3: Creditworthiness

As in Table 2, this table reports estimates of equations 1 and 2. *Immigrant* is an indicator for whether the consumer's SSN was assigned at age 21 or older; *SSN Age* is the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group). The outcomes in Panel A are average credit score, measured by VantageScore, by ages 30 (columns 1 and 2) and 40 (columns 3 and 4). These Panel A outcomes are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). The outcomes in Panel B are the probability of prime or higher credit score, measured by a VantageScore of 660 or higher, by age 30 (columns 1 and 2) and age 40 (columns 3 and 4), where consumers without a credit score are given a value of zero (for consumers not observed at a given age, we use the score at the oldest age observed before that age to measure whether their score is prime or higher). The units in Panel A are credit score points, and the units in Panel B are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Average credit scores

Dep Var: Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	26.7***	17.2***	24.6***	14.2***
	(0.4)	(0.6)	(0.4)	(0.5)
SSN Age		2.3***		2.4***
		(0.1)		(0.1)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.135	0.135	0.124	0.124
N	6,005,430	6,005,430	6,250,733	6,250,733
Mean, SSN Age <21	624.2	624.2	658.1	658.1

Panel B: Probability of prime or higher credit score

Dep Var: Prime or Higher by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	4.61***	6.82***	11.24***	7.04***
	(0.17)	(0.25)	(0.17)	(0.22)
SSN Age		-0.51***		0.98***
		(0.05)		(0.04)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.102	0.102	0.098	0.098
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	37.63	37.63	49.33	49.33

Table 4: Credit access

As in Table 2, this table reports estimates of equations 1 and 2. *Immigrant* is an indicator for whether the consumer's SSN was assigned at age 21 or older; *SSN Age* is the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group). The outcomes in Panel A are age at first auto loan, age at first mortgage, and age at first credit card (for consumers who do not access a product by the end of our data, we assign them a value that corresponds to accessing that product in 2025, one year after the end of our data). The outcomes in Panel B are the probability of having ever held an auto loan by age 37, held a mortgage by age 37, or held a credit card by age 37. The units in Panel A are years of age and the units in Panel B are percentage points. Standard errors are clustered by birth year × the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Panel A: Age of credit products first taken out

Dep Var: Age at First...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	4.96***	2.27***	2.26***	0.40*	4.80***	1.79***
	(0.05)	(0.05)	(0.04)	(0.04)	(0.02)	(0.03)
SSN Age		0.63***		0.43***		0.70***
		(0.01)		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.083	0.084	0.082	0.083	0.082	0.086
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	29.40	29.40	36.49	36.49	22.99	22.99

Panel B: Probability of having held a credit product by age 37

Dep. Var: By Age 37, has ...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66***	-8.69***	-8.91***	-1.05***	-1.35***	-0.37***
	(0.18)	(0.23)	(0.17)	(0.23)	(0.08)	(0.11)
SSN Age		-1.62***		-1.83***		-0.23***
		(0.04)		(0.04)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.041	0.042	0.063	0.063	0.024	0.024
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Table 5: Cumulative years of delayed access to credit from paired cohorts

This table presents estimates of the cumulative years of delayed access to credit due to immigration occurring a single year later, obtained from the paired cohort OLS regression strategy. Each column shows the results of a separate regression on a different outcome, and each row shows estimates for a different event time coefficient. The row "C" indicates the event time year in which the control group (which immigrated 1 year before the treatment group, i.e., $C \equiv T-1$) is assigned an SSN while "T" is the event time year of the focal (or treatment) cohort's immigration. The specification includes all year lags from event time C through to $T+13$, thirteen years after the treatment cohort's immigration. The omitted category is $T-2$. 95% confidence intervals are shown in [.], which are calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations. *$p < .05$; **$p < .01$; ***$p < .005$.

Dep. Var: Any ...	Credit Score (1)	Credit Product (2)	Auto Loan (3)	Mortgage (4)	Credit Card (5)
C	-0.076***	-0.243***	-0.03***	-0.005***	-0.227***
	[-0.080, -0.072]	[-0.251, -0.235]	[-0.034, -0.026]	[-0.006, -0.003]	[-0.235, -0.218]
T	-0.390***	-0.476***	-0.09***	-0.021***	-0.448***
	[-0.403, -0.377]	[-0.490, -0.463]	[-0.100, -0.080]	[-0.027, -0.015]	[-0.463, -0.434]
$T+5$	-0.716***	-0.684***	-0.327***	-0.160***	-0.656***
	[-0.745, -0.686]	[-0.717, -0.653]	[-0.347, -0.308]	[-0.195, -0.126]	[-0.689, -0.625]
$T+10$	-0.778***	-0.725***	-0.443***	-0.283***	-0.692***
	[-0.813, -0.741]	[-0.762, -0.689]	[-0.473, -0.412]	[-0.338, -0.228]	[-0.729, -0.656]
N Paired Cohorts	68	68	68	68	68
N Consumers	463,542	463,542	463,542	463,542	463,542
N	7,416,672	7,416,672	7,416,672	7,416,672	7,416,672

Table 6: American Community Survey: income

This table uses public data from the 5-year American Community Survey (ACS) waves 2009, 2013, 2018, and 2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each column shows results of a separate cross-sectional regression. All columns include an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns also include *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. The outcome is Log (Income), the log of household income that is winsorized at 99.9th percentile. The baseline mean is the household income in dollars for non-immigrants. All regressions also include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, and survey year. Units are percentage points. Heteroskedasticity-robust standard errors. $^{+}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .005$.

	Log (Income)							
	2006 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-17.66***	-11.79***	-17.41***	-10.18***	-14.51***	-8.55***	-8.08***	-3.03***
	(0.36)	(0.67)	(0.31)	(0.56)	(0.28)	(0.53)	(0.34)	(0.64)
Immigration Age		-1.49***		-1.69***		-1.27***		-1.06***
		(0.14)		(0.11)		(0.10)		(0.12)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R^2	0.161	0.161	0.196	0.196	0.239	0.239	0.228	0.228
Mean Income, Immigration Age <21	\$71,088	\$71,088	\$78,339	\$78,339	\$98,660	\$98,660	\$134,099	\$134,099

Table 7: Survey of Consumer Finances (2022): debt attitudes

This table uses public data from the 2022 Survey of Consumer Finances (SCF), keeping birth years 1972 to 2000 such that respondents are aged between 22 and 49. Odd-numbered columns are estimates from the cross-sectional OLS regression that includes an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns add *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. See Appendix A.2 for details of the survey variables used. The units are percentage points, except for columns 7 and 8 where the estimates are multiplied by 100 to be on a similar scale to percentage points (this outcome takes values -1, 0, and +1). All regressions include a control for normal income (winsorized at the 99th percentile), and fixed effects for the birth year of the consumer and fixed effects for other demographic control variables: male, race, Hispanic, spouse, marital status, household size, number of children, and education levels. Columns 3 to 6 also include a control for the binary auto loan debt attitude variable, where a value of 1 is considering borrowing for auto is all right. Heteroskedasticity-robust standard errors. Missing data in the SCF are imputed five times using a multiple imputation technique, storing data in five "implicates". We run a separate regression on each of the five implicates and follow the SCF's recommended multiply-imputed variance estimation technique for combining standard errors. $^{+}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .005$.

Dep Var:	Auto Debt Attitude		Any Auto Debt		Any Auto Debt		General Debt Attitude	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-15.41***	-23.02***	-4.27	-11.14	-0.21	-5.10	-8.96	-15.94
	(4.58)	(7.25)	(5.08)	(8.27)	(5.01)	(7.87)	(7.92)	(13.71)
Immigration Age		0.98		0.88		0.63		0.89
		(0.62)		(0.76)		(0.69)		(1.19)
Auto Debt Attitude					26.34***	26.24***		
					(3.23)	(3.24)		
Demographic Controls	X	X	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.063	0.106	0.106	0.026	0.026
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14	3.80	3.80

Supplemental Appendix: Immigration and Credit in America

by J. Anthony Cookson, Benedict Guttman-Kenney and William Mullins

A. Supplemental Details On Data

A.1 Additional Details On Credit Reporting Data Construction

We start constructing our data by taking all consumers observed between July 2000 and September 2023 in the University of Chicago Booth TransUnion Consumer Credit Panel, "BTCCP" (TransUnion, 2025). We first queried the BTCCP for this project in September 2023, however, because the data are regularly updated, we observe credit outcomes through to December 2024. We focus the sample on consumers who had at least one tradeline at any point between July 2000 and September 2023. This restriction drops fragmented credit reports, such as those with only credit inquiries. We retain consumers with valid birth dates between 1951 and 2004 because we need birth dates to calculate the year of immigration. "Consumers" without birth dates are more likely to be credit reports in which debts are fragmented across multiple consumer identifiers; Gibbs et al. (2025) recommends dropping these observations. We start in 1951 for two reasons. First, prior to 1951, there are spikes around particular birth dates, suggesting that some older birth dates may be less accurate in this period. Second, the timing of deaths is not well measured in credit reporting data (Gibbs et al., 2025). Focusing the sample on relatively recent birth years limits this issue. These restrictions produce an anonymous sample of 31.87 million consumers, representative of 318.7 million consumers.

To estimate the year of SSN assignment, $SSN\ Year$, we construct a lookup table that maps the first five digits of the SSN into the first year of SSN assignment, $SSN\ Year$, for all SSNs assigned before 2012. Some 5-digit SSN sequences are assigned over a 2 or 3 year period, while others are assigned in a single year. In all cases, we take as the year of SSN assignment the *first* year the 5-digit sequence was assigned. This means that a subset of our consumers will, in reality, receive a SSN one or two years later than they appear to in our data. We sent TransUnion the list of all 31.87 million consumers in the full sample, together with this lookup table. TransUnion then matched the consumer list and the lookup table to their underlying data, which includes consumers' SSNs, returning a dataset with the year of SSN assignment, an indicator for whether a consumer had any SSN in their data, and the BTCCP consumer identifier. This procedure guarantees that we never observe nor can we infer SSNs for any consumers in the BTCCP.[1]

Panel A of Table A1 describes how we arrive at our sample. After removing the 21% of consumer identifiers without SSN information, we have 25.2 million consumers. Requiring SSN information appears to remove fragmented credit records with younger ages and credit reports that do not persist over time. It is common to focus research on credit reports with SSNs, most notably, the widely-used Federal Reserve Bank of New York's Consumer Credit Panel *only* contains information on consumers with SSNs (Gibbs et al., 2025). More fundamentally, our immigrant classification relies on SSN information in a consumer's TransUnion file. Without a SSN, we cannot establish a consumer's year of SSN assignment. Fragmentation will be less common in files with SSNs and birth dates because these identifiers are used to consolidate files. In our sample, each SSN is unique; thus, cases with fragmented files may slightly undercount the debt of those consumers.

Without further refinement, this procedure is noisy because all nine digits of SSNs were randomly assigned after mid-2011. Some of these later-assigned SSNs fall into blocks in our lookup table, erroneously assigning them to earlier SSN Years. To avoid such misclassification, we remove any consumers who have $SSNAge < 0$, and any consumers with birth years of 1988 or later, as they are too young to enter the data or to be classified as immigrants by 2011, or if their credit report was established before age 15, indicating a lower quality birth date.

In an earlier version of our paper, we found our results to be robust to including an additional sample restriction that consumers have a credit report by 2011 which excludes an additional 3% of consumers. With this restriction, the positive selection of immigrants remains, we estimate that immigrants have significantly higher mean credit score at ages 30 and 40 by 27.0 (compared to 26.7 in this paper without the exclusion) and 26.7 (compared to 24.6 without the exclusion) points, respectively, compared to non-immigrants after controlling for birth year and first Zip5 fixed effects. The estimates for credit access are also similar. For example, by age 37, immigrants are 12.99 (compared to 15.66) percentage points significantly less likely to have ever held an auto loan and 7.54 (compared to 8.91) percentage points significantly less likely to have held a mortgage, compared to non-immigrants after controlling for birth year and first ZIP5 fixed effects.

Panel A of Table A1 shows that the resulting *"Clean Sample"* has 19.02 million consumers, 2.23 million of whom we classify as immigrants, based on a SSN Age of 21+. The 11.7 percent of

[1] This classification depends on knowledge of the first five digits of a consumer's SSN. Bernstein et al. (2025) use Infutor data, which was originally built from consumer credit reporting data. However, since the 1999 Gramm-Leach-Bliley Act, the bureaus cannot sell address lists to third parties, forcing Infutor and its competitors to rely on other data sources.

immigrants in this clean sample approximates the 10.2 percent of consumers in the ACS, applying the same birth year restrictions and also only classifying immigrants if their year of immigration was when they are aged 21 or older. We then apply additional restrictions to arrive at the *"Entrant Sample"* used for our paper's analysis. These restrictions are that consumers have a SSN Age of < 30 and birth years from 1975 to 1987 for the reasons described in the main paper. Within this sample, there are 102 immigration combinations of birth years and SSN Ages. As shown by the counts of consumers by SSN Age in panel B of Appendix Table A1, each SSN Age group in this entrant sample has many consumers, ranging from 57,114 at SSN Age 21 to 22,399 at SSN Age 29. Panel B of Appendix Table A1 also shows that, of the consumers with birth years 1975 to 1987, there are 55,704 consumers assigned a SSN Age of 30+, who we do not include in the Entrant Sample. This means that our SSN Ages 21 to 29 cover 87% of legal immigrants who arrive during adulthood in our data for those birth years.

Further validation shows a close correspondence between our estimates and the ACS estimates of immigrants. In particular, panel A of Appendix Figure A2 shows how the distribution of immigrants by birth year compares in our classification, using SSN Age of 21+ in the BTCCP, to the ACS, using age of immigration of 21+. We see that from birth years 1970 onwards – we use cohorts born in 1975 and onwards in our analysis – our estimate for the percentage of immigrants within each birth year closely aligns with the ACS estimates.[2] Lower immigration rates may be expected for our data as we only capture legal immigrants with SSNs, whereas the ACS estimates also include illegal immigrants and legal immigrants without SSNs. However, immigration rates may be higher in our data given that immigrants are under-reported in the ACS, as respondents may not want to disclose their immigrant status (Brown et al., 2023). In panel B of Appendix Figure A2, we see that using the Age at SSN Assignment in our data gives a similar distribution of adult age of immigration to the distribution found in the ACS. Both panels suggest that our immigrant classification matches the characteristics of the overall immigrant population *despite* focusing on consumers who have SSNs and whose credit reports are not fragmented. Finally, in panel C of Appendix Figure A2 we show that the immigrant classification delivers a close match to the distribution of immigrants across states found in the ACS. Finally, Appendix Figure A5 shows that the consumers we classify as immigrants are more geographically mobile than non-immigrants, as would be expected with

[2]Before birth year 1969 our classification appears to overestimate the share of immigrants. One reason for this is non-immigrants obtaining SSNs after age 21, but this practice was reduced by the Social Security Amendments of 1972 (P.L. 92-603), which authorized the provision of SSNs to children at the time they first entered school at around age 6 (https://www.ssa.gov/history/ssn/ssnchron.html) reducing mis-classification for cohorts born in 1967 onwards.

reliable classification. These validation exercises, together with similar validations in the existing literature using the age of SSN assignment (Bernstein et al., 2025, Klopfer and Miller, 2024), suggest that our strategy to classify immigrants is reliable.

A.2 Additional Details On The Survey of Consumer Finances (2022)

We use the 2022 wave of the SCF as this is the only wave that includes information on whether a consumer is an immigrant and the number of years since immigration. When using the SCF, we apply similar birth year restrictions to our credit reporting data, and doing so reduces the SCF sample size from 4,595 to 1,813 respondents, of whom we classify 213 as immigrants based on an age of immigration of 21+. We study respondents in their twenties, thirties, and forties in the 2022 survey year (corresponding to birth years 1972 to 2000). We use this restriction because consumers in their fifties and older are expected to have limited credit demand irrespective of immigration status or credit supply. Given the small survey sample size, we retain respondents with ages of immigration older than 30, in contrast to the restriction used in the credit reporting data and the ACS.

When analyzing the SCF, we use cross-sectional regressions of the same overall form as Equations 1 and 2, with a few minor changes. Rather than using *SSN Age*, we directly observe the age of immigration in the survey, and therefore $\mathbb{I}\{Immigrant\}_i$ measures whether an a respondent immigrated at age 21 or later in life, and instead of $SSN\ Age_i$ we have $Immigrant\ Age_i$, a variable for each year of immigrating at age 21 or later in life. Given the small survey sample size, we increase the precision of our regression estimates by including demographic controls for normal income (winsorized at the 99$^{\text{th}}$ percentile) and fixed effects for birth year ($\mu_{b(i)}$), male, race, hispanic, spouse, marital status, household size, number of children, and education levels, but cannot include geographic fixed effects ($\eta_{g(i)}$) as these are not in the data. The SCF is conditional on a consumer being in the United States when surveyed in 2022 and, therefore our results are conditional on not emigrating. We use the SCF's recommended survey weights in our regressions to correct for the over-sampling of high-wealth consumers in the survey's design, and we use heteroskedasticity-robust standard errors and follow the SCF's recommended multiply-imputed variance estimation technique to account for missing data.

In all of the regression in Table 7 and Appendix Table A8, the age of immigration is constructed from the survey variable x6906 (years lived in the United States) and from variable x8022 (age). The indicator for immigration status (*Immigrant*) takes a value of one if the consumer's age of immigration was at 21 or older, and zero otherwise. The variable *Immigration Age*, the consumer's age at immigration, pools all consumers with ages of immigration at 20 or younger at a value of 20.

In Table 7 columns 1 and 2 of Panel A and also columns 5 and 6 of Panel B, the outcome in is whether a consumer has any auto vehicle (constructed from variable x2201). The outcome in columns 3 and 4 of Panel A and also columns 7 and 8 of Panel B is whether a consumer has any outstanding auto loan debt or lease on any of their vehicles (constructed from the auto financing and leasing variables x2102, x2208, x2308, x2408, x7157). The mean of this variable may appear to be lower than one might expect, given that most car purchases in the United States are financed with debt (see cites in Gibbs et al., 2025), however, this makes sense because some consumers will have previously taken out auto loans but have paid this debt off by the time of being surveyed. The outcome in columns 5 and 6 of Panel A is whether a consumer is a homeowner (i.e., not a renter, constructed from variables x701 and x601). The outcome in columns 7 and 8 of Panel A is whether a consumer has any outstanding mortgage debt (constructed from x723). The outcome in columns 9 and 10 of Panel A is the total credit card limits across all cards (constructed from variable x414, and winsorized at the 99th percentile).

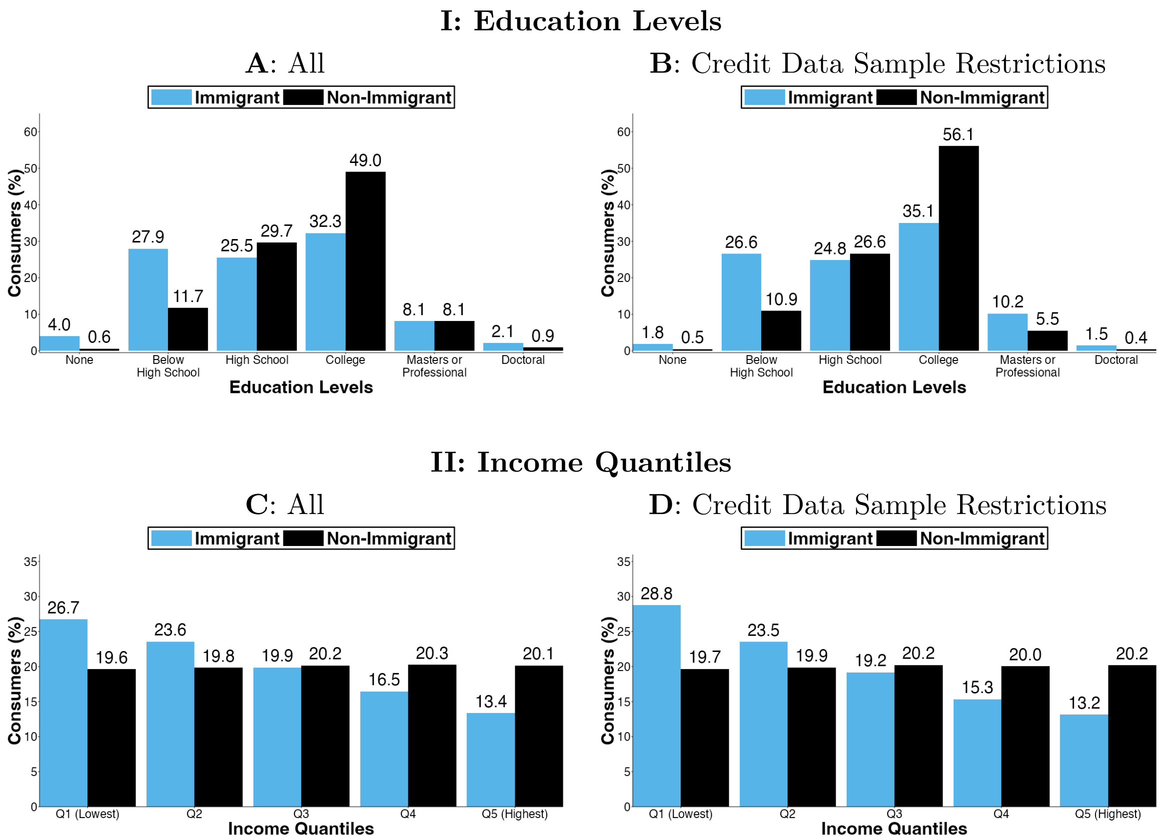
In Panel B of Table 7, the outcome in columns 7 and 8 is "Now I would like to ask you some questions about how you feel about credit. In general, do you think it is a good idea or a bad idea for people to buy things by borrowing or on credit?" (constructed from variable x401), with the variable having three potential values: 1 if they regard it as a good idea, 0 if good in some ways, bad in others, and -1 if they regard it as a bad idea. In this table we multiply estimates by 100 to be on a similar scale to other estimates. In Panel B of Appendix Table A8, we decompose this variable into binary measures. In columns 1 and 2 of Appendix Table A8, this outcome takes a value of 1 if they regard it as a good idea, and zero otherwise. In columns 3 and 4 of Appendix Table A8, this outcome takes a value of 1 if they regard it as a bad idea, and zero otherwise.

The outcome (constructed from variable x405) in columns 1-2 of Panel B of Table 7, and also used as a control variable *Auto Debt Attitude* in columns 3 to 6 of this same panel, is a binary variable for whether they feel it is all right to borrow money to finance the purchase of a car. This is following the prompt "people have many different reasons for borrowing money which they pay back over a period of time, please tell me whether you feel it is all right for someone like yourself to borrow money...", taking a value of 1 if they respond yes and a value of 0 if they respond no.

The outcomes in columns 5 to 10 of Appendix Table A8 Panel B are all binary variables responding to different scenarios following the prompt “people have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...”, taking a value of 1 if they respond yes and a value of 0 if they respond no.The outcome in columns 5-6 (constructed from variable x403) is whether they feel it is all right for someone like yourself to borrow money to cover living expenses when income is cut. The outcome in columns 7-8 (constructed from variable x402) is whether they feel it is all right for someone like yourself to borrow money to cover the expenses of a vacation trip. The outcome in columns 9-10 (constructed from variable x406) is whether they feel it is all right for someone like yourself to borrow money to finance educational expenses.

B. Supplemental Figures and Tables

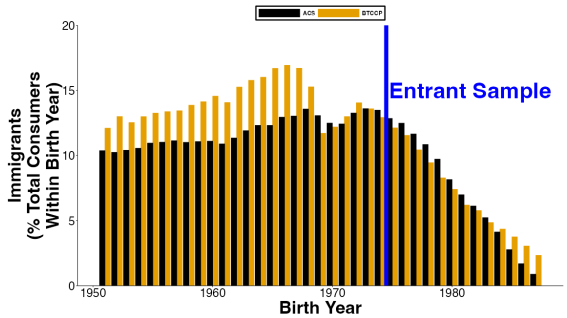
Figure A1: Distribution of Education and Income of Immigrants Compared to Non-Immigrants in the American Community Survey (ACS)



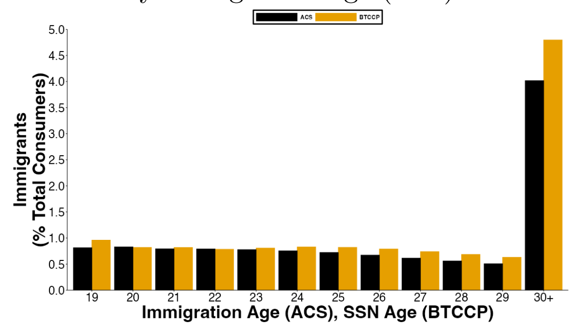
This figure summarizes data computed from the 5-year 2010 American Community Survey (ACS). Panels A and C includes all consumers with positive income and who are aged between 18 and 80. Panels B and D also restrict to birth years 1975 to 1987 and excludes immigrants who arrived from age 30 or older to create a sample that is comparable to the Entrant sample that we use in our analysis of credit data. In all panels, immigrants are classified as those whose age of immigration is at age 21 or older, with everyone else classified as non-immigrants. In each panel, each black bar shows the baseline means for the non-immigrant group. In each panel, each blue bar shows the baseline mean added to the estimates from regressions on an indicator for immigration age, where the regressions also control for birth year, geographic area (Consistent Public Use Microdata Areas), and survey year. The outcomes in Panels A and B are indicators for each level of education. The outcomes in Panels C and D are indicators for each quintile of household income.

Figure A2: Immigrant classification in our data (BTCCP) versus the American Community Survey (ACS)

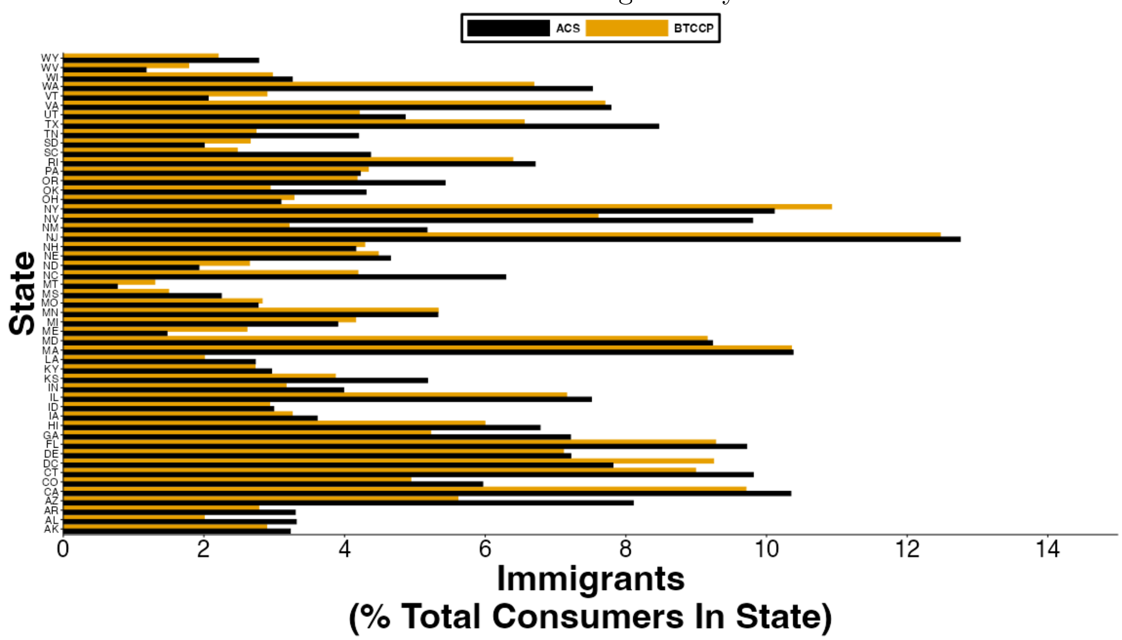
A: Distribution of immigrants by birth year



B: Distribution of immigrants by immigration age (19+)



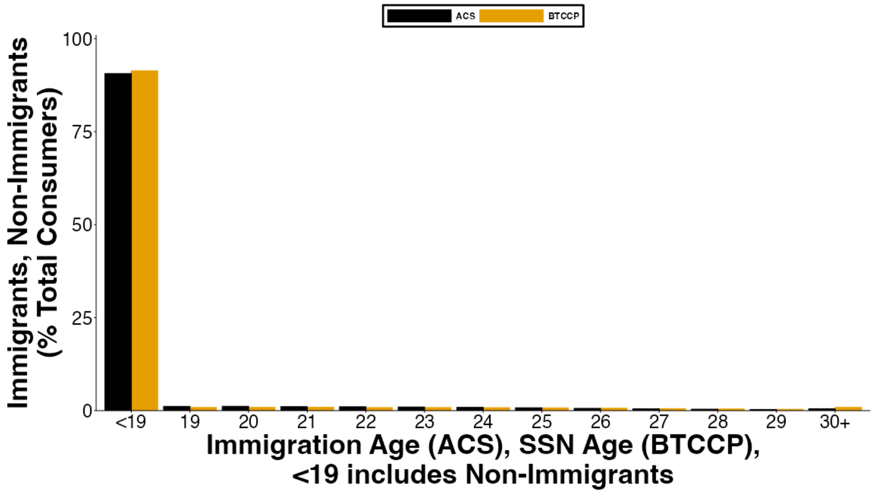
C: Distribution of immigrants by state



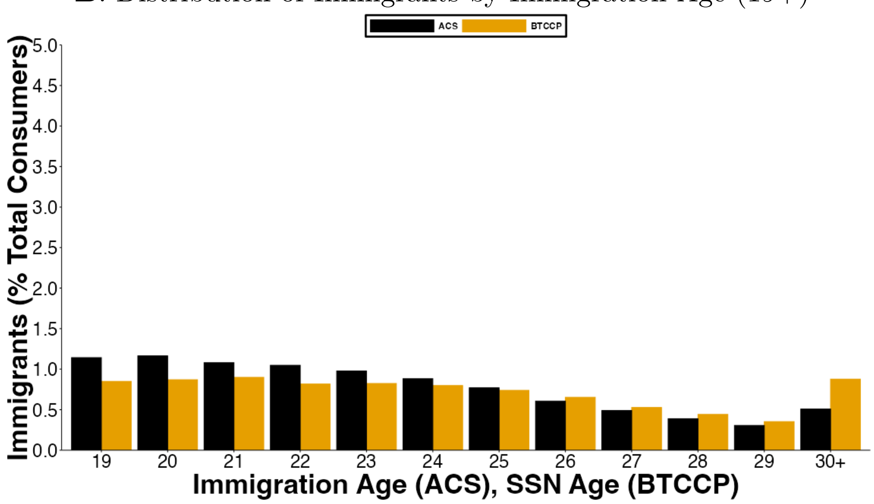
This figure compares our classification of immigrants in the Booth TransUnion Consumer Credit Panel (BTCCP) to comparable statistics computed from the American Community Survey (ACS). Panel A presents the share of immigrants in each birth year in the ACS (black) versus BTCCP (yellow). In Panel A, immigrants are defined as SSN at age 21+ in the BTCCP and if their age of immigration is 21+ in the ACS. The birth years to the right of the blue vertical line are the "Entrant Sample" of birth years 1975 to 1987 that are used for our analysis. Panel B presents the share of each sample by age at immigration, non-immigrants or those that immigrate at age 18 or younger are included in the denominator but the bar is excluded from this figure to ease presentation, it is 88.15% for the ACS and 86.52% for the BTCCP.

Figure A3: Immigrant classification in our data (BTCCP) versus the American Community Survey (ACS) for entrant sample, with SSN Age 30+ added

A: Distribution of Non-Immigrants and Immigrants by Immigration Age



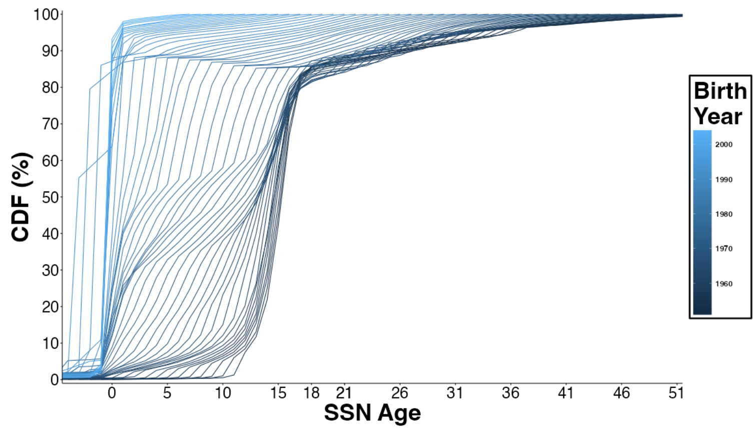
B: Distribution of Immigrants by Immigration Age (19+)



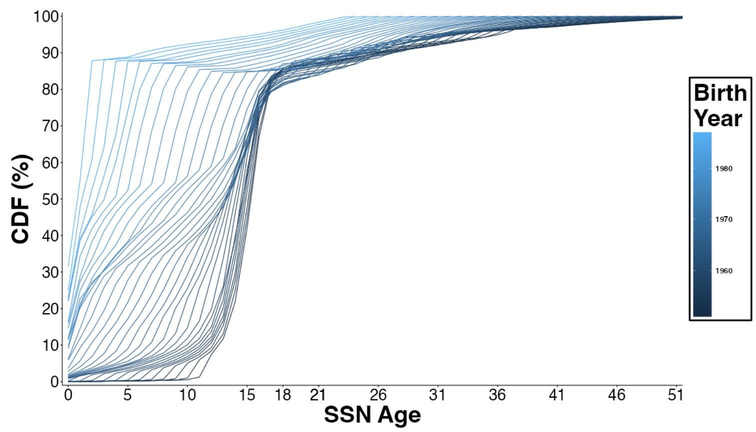
This figure compares our classification of immigrants in the BTCCP Entrant Sample to comparable statistics computed from the American Community Survey (ACS). The figure shows bars for immigrants with Age of Immigration (ACS) / SSN Age of 30+, these are consumes who are not included in the Entrant Sample used for analysis.Panel A presents the share of immigrants by immigration age, where < 19 includes non-immigrants or those that immigrate at age 18 or younger. Panel B presents a zoomed in version of Panel A that only shows the bars for the subset of immigration ages from 19 to 30+.

Figure A4: Distributions of SSN Age by birth year

A: Before cleaning sample

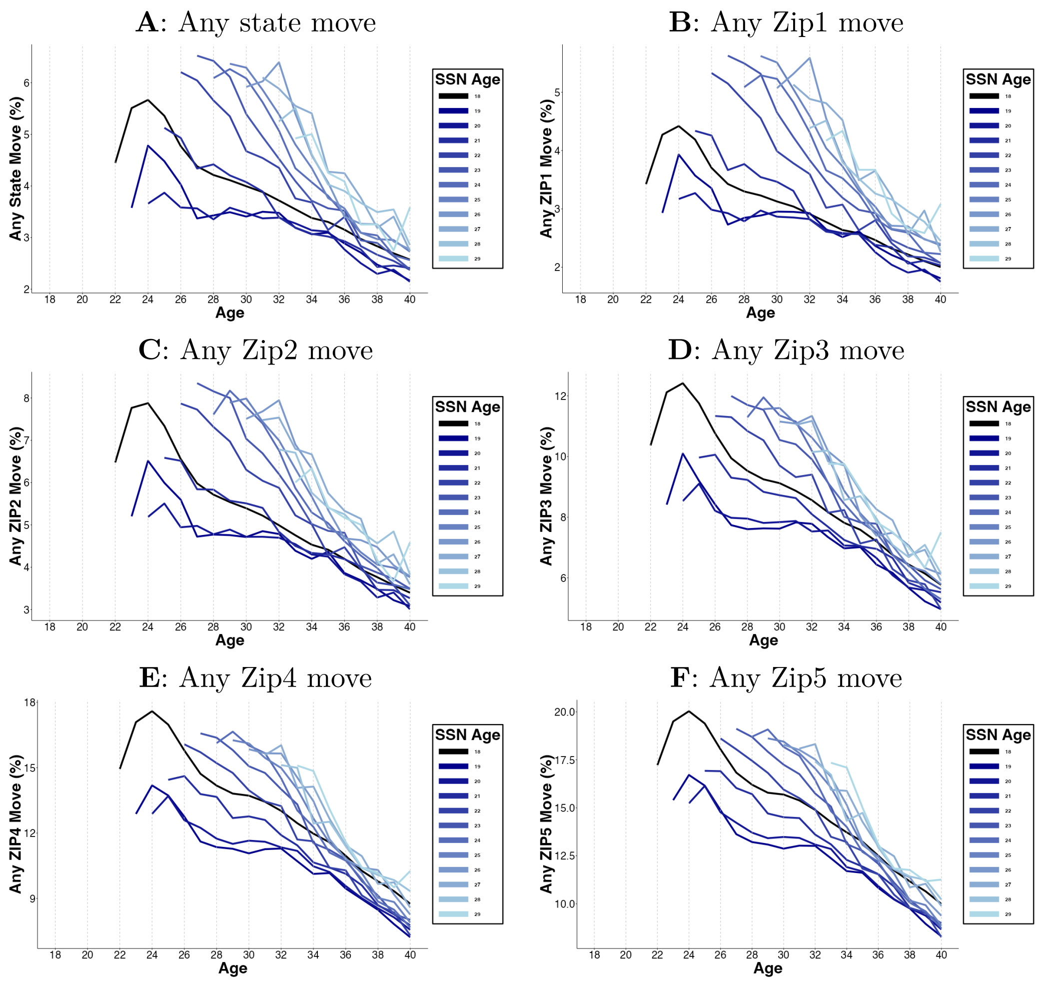


B: Clean sample



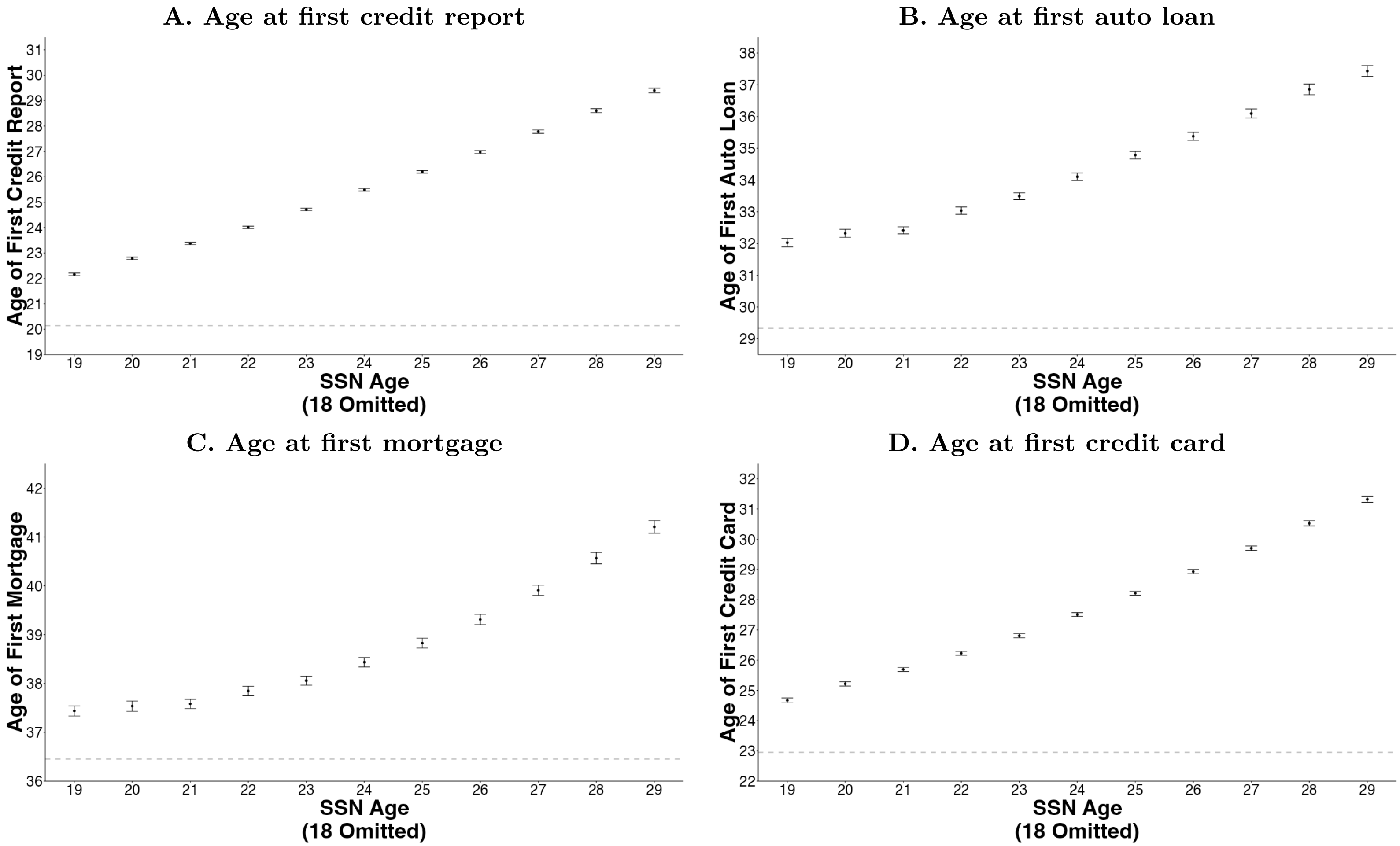
These panels show the CDFs of SSN Age for each birth year. Panel A shows data before cleaning (only removing consumers without SSNs). Panel B restricts to consumers after cleaning the data. These patterns of SSN Ages by birth years are consistent with Klopfer and Miller (2024) using administrative Social Security Administration data.

Figure A5: Lifecycle of geographic mobility by SSN Age cohorts



For each SSN Age cohort, this figure shows the share of consumers at each age who move state (Panel A), ZIP1 (Panel B), ZIP2 (Panel C), ZIP3 (Panel D), ZIP4 (Panel E), and ZIP5 (Panel F). The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSNAge$ is indicated by the circles on each line. This figure uses data for birth years from 1982 to 1987. Consumers with the birth years 1985, 1986, and 1987, are not observed for ages 40, 39 to 40, and 38 to 40, respectively, by the end of our data in 2024.

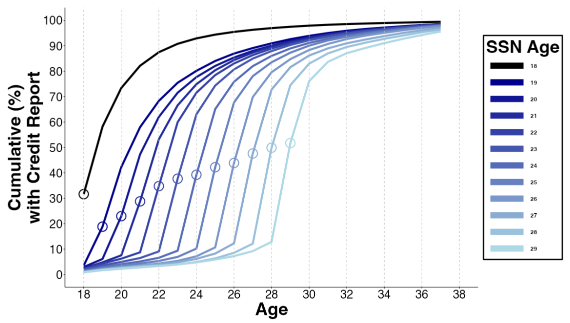
Figure A6: Age at first credit product by SSN Age



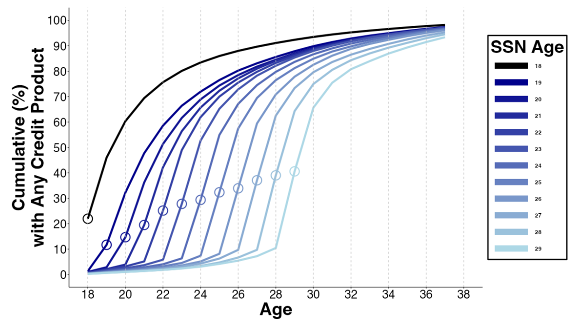
This figure presents estimates and 95% confidence intervals for age at first credit report (Panel A), age at first auto loan (Panel B), age at first mortgage (Panel C), and age at first credit card) separately by SSN Age cohorts. The estimates are constructed from an individual-level regression of each outcome on SSN Age indicators, Birth Year fixed effects, and fixed effects for consumers' first observed ZIP code. The baseline mean for the omitted category (SSN Age 18 or lower) is indicated by the dashed gray line. In these regressions, for consumers that do not access a product by the end of our data, we assign them a value that corresponds to accessing that product in 2025, one year after the end of our data. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code.

Figure A7: Additional credit outcomes by SSN Age cohorts

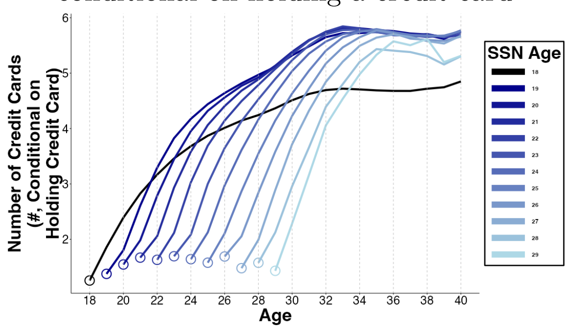
A: Any credit report



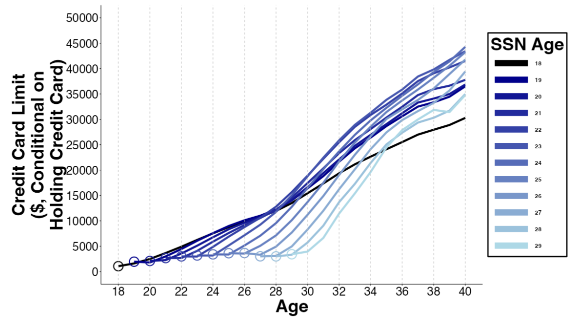
B: Any credit product



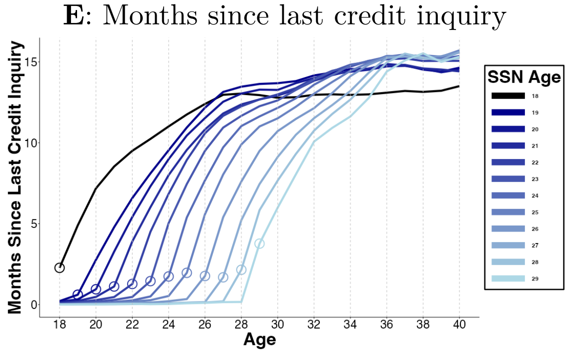
C: Number of credit cards, conditional on holding a credit card



D: Credit card limits, conditional on holding a credit card



E: Months since last credit inquiry

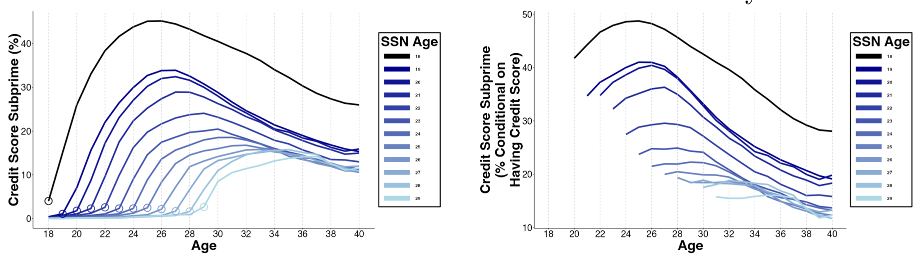


For each SSN Age cohort, this figure presents the evolution of age of first credit report (Panel A) and age of first credit product (Panel B) by age, using data for birth years from 1975 to 1987. These averages are conditional on having a credit score. Because Panels A and B are cumulative charts we end them at age 37, whereas for the other panels the estimates for ages 38-40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for ages 38 to 40, 39 to 40, and 40 respectively by the end of our data in 2024. Panels C, D, and E use data for birth years from 1982 to 1987 to show the mean number of credit cards, conditional on holding a card (Panel D), the mean credit card limits, conditional on holding a credit card (Panel E), and the mean number of months since last credit inquiry (Panel F) at each age for consumers of each SSN Age. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSNAge$ is indicated by the circles on each line.

Figure A8: Lifecycle of credit scores by SSN Age cohorts

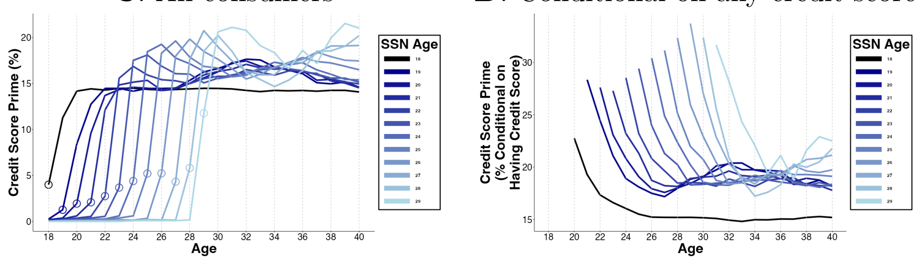
I: Any subprime credit score

A: All consumers **B**: Conditional on any credit score



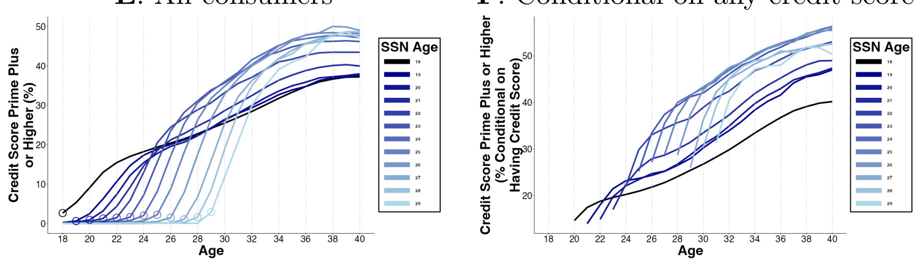
II: Any prime credit score

C: All consumers **D**: Conditional on any credit score



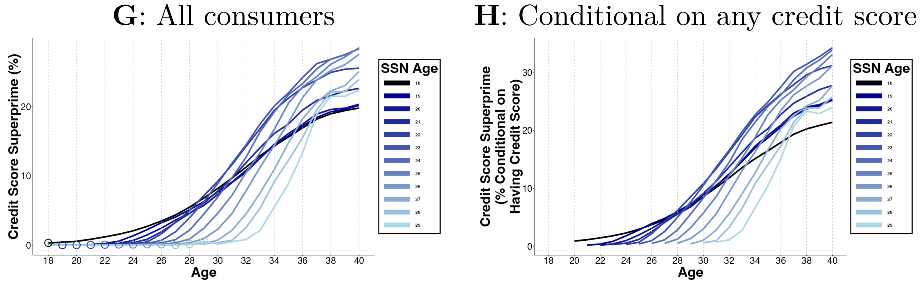
III: Any prime plus or superprime credit score

E: All consumers **F**: Conditional on any credit score



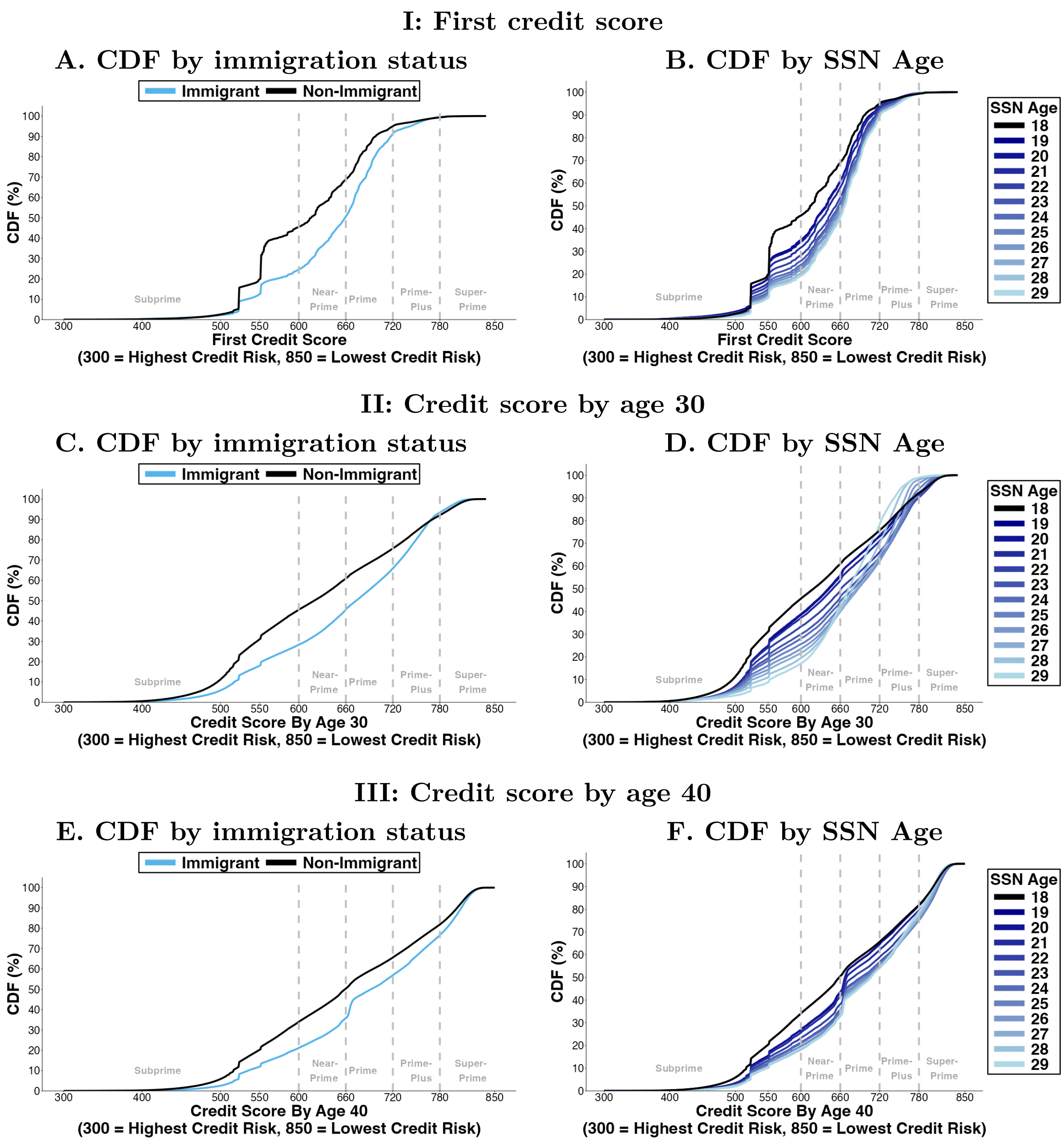
IV: Any superprime credit score

G: All consumers **H**: Conditional on any credit score



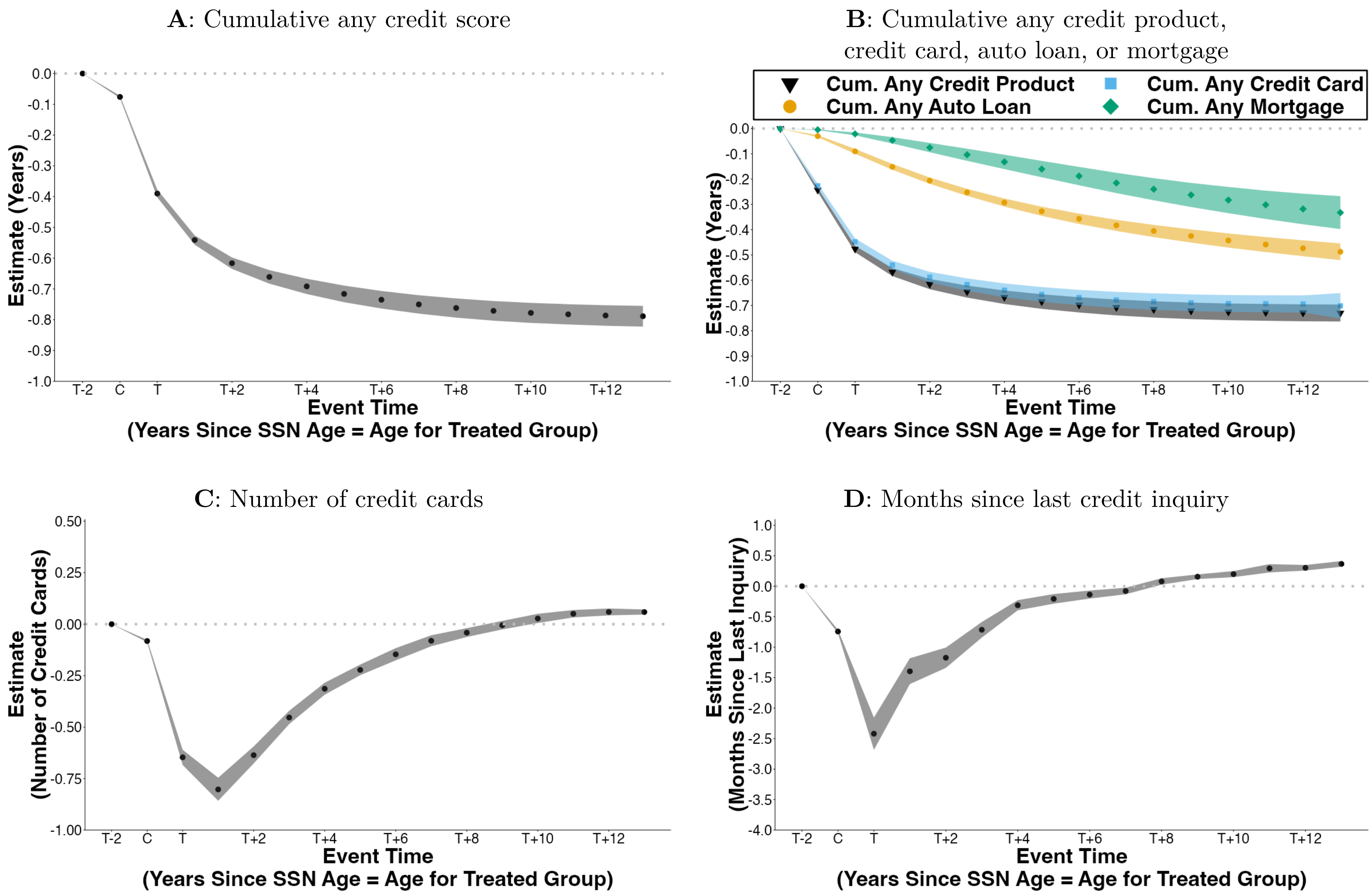
For each SSN Age cohort, this figure presents the evolution of credit scores by age. In all panels, credit scores are measured by VantageScore. In Panels A and B, Subprime Credit Score is VantageScore below 600. In Panels C and D, Prime Credit Score is VantageScore 661 to 719. In Panels E and F, Prime Plus or Superprime Credit Score is VantageScore 720 or higher. In Panels G and H, Superprime Credit Score is VantageScore of 780 or higher. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSNAge$ is indicated by the circles on each line. This uses data for birth years from 1982 to 1987.

Figure A9: Distribution of credit scores



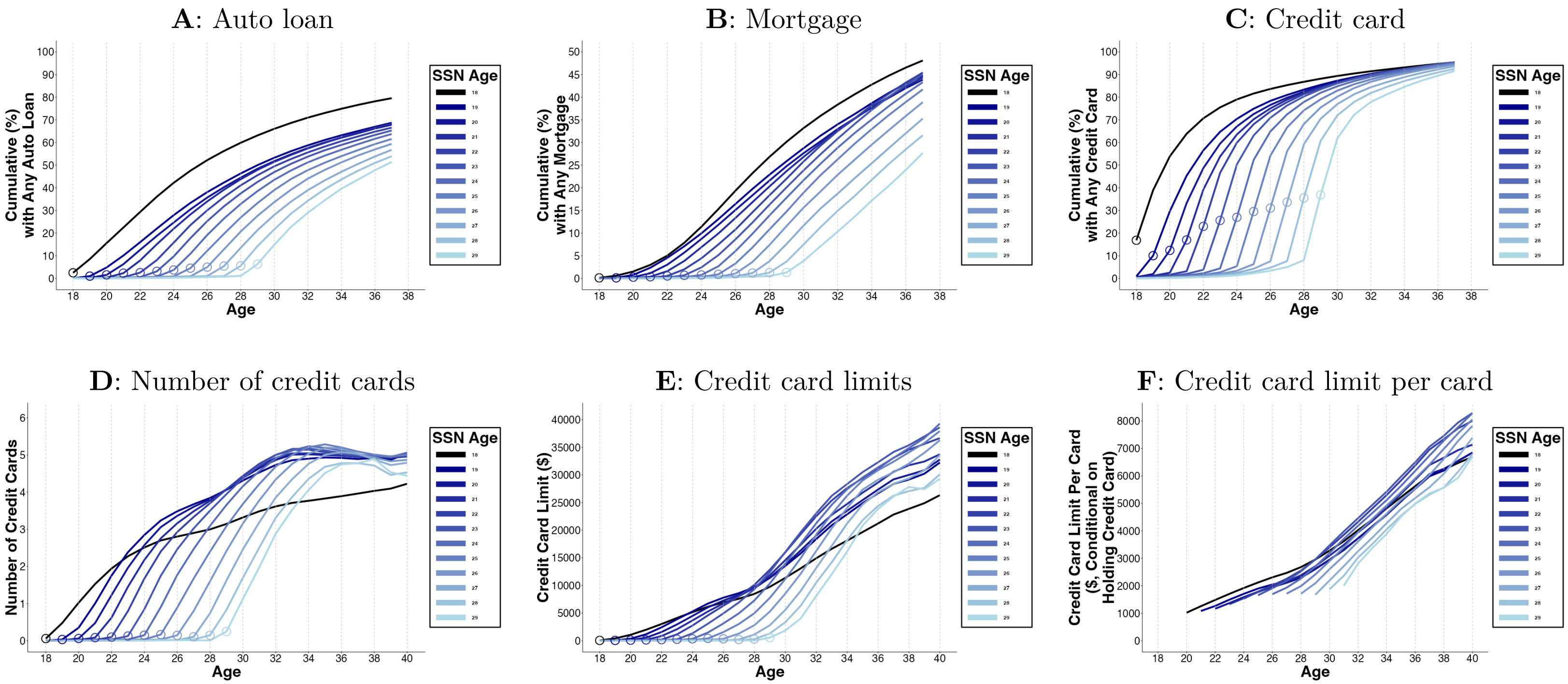
Panels A and B show the CDFs of the first credit score observed for a consumer at any point in our data. Panels C and D show CDFs of credit score by Age 30. Panels E and F show CDFs of credit score by Age 40. Panels A, C, and E show CDFs by immigration status, where immigrants are defined as SSN Age 21 to 29. Panels B, D, and F show CDFs by SSN Ages, where SSN Age 18 includes consumers that are assigned an SSN at Age 18 or younger. The outcomes in Panels C to F are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). Panels A and B are also conditional on having a non-missing credit score, and we also drop consumers with credit scores that are first observed in July 2000, where our data begins, as credit scores that are first observed at this point-in-time may not necessarily reflect a consumer's first credit score, just a consumer's first score in our data.

Figure A10: Paired cohorts: dynamics of credit access



This figure presents dynamic estimates for differences in credit access (of different types) between a cohort with SSN Age = s and a cohort with SSN Age = $s-1$ matched at the same age. The differences are presented in event time where C is the year when age equals SSN Age for the $s-1$ cohort and T is the same for the s cohort: first credit product or credit card (Panel A), auto loans and home mortgages (Panel B), and credit card limits (Panel C). The shaded areas indicate 95% confidence intervals, calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations.

Figure A11: Lifecycle of credit access by SSN Age cohorts, restricting sample to consumers still observed in 2024



This figure restricts the sample to consumers with a non-missing credit score in 2024. For each SSN Age cohort, this figure shows the cumulative share of consumers at each age who have ever had an auto loan (Panel A), a mortgage (Panel B), or a credit card (Panel C). Panels A, B, and C use data for birth years from 1975 to 1987. Because Panels A, B, and C are cumulative charts we end them at age 37, whereas for the other panels the estimates for ages 38-40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for 40, 39 to 40, and 38 to 40, respectively, by the end of our data in 2024. Panels D, E, and F use data for birth years from 1982 to 1987 to show the mean number of credit cards (Panel D), the mean credit card limits (Panel E), and the mean credit card limit per card (Panel F) at each age for consumers of each SSN Age. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSNAge$ is indicated by the circles on each line.

Table A1: Data Sample

Panel A presents an observation funnel that details how we obtain our final sample starting from the full BTCCP 10% sample. As each sample restriction is applied, the table details how the number of consumers changes, until we obtain the dataset used for our analysis: the *Entrant Sample*. Panel B presents observation counts for the "Clean Sample" (indicated in Table A1) by SSN Age in column 1. Column 2 presents similar counts for consumers whose SSN Assignment year falls between 2000 and 2012, and thus, fall in our entrant sample. Our entrant sample additionally drops the 55,704 consumers who have SSN Age 30+, these consumers are included in Panel B of this table for completeness.

Panel A: Sample construction

TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
Clean Sample	19,019,191
... SSN Age <21	16,793,939
... SSN Age 21+	2,225,252 (11.70%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
Entrant Sample (SSN Age < 30)	6,299,287
... SSN Age < 21	5,914,891
... SSN Age 21-29	384,396 (6.10%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

Panel B: Counts of consumers by SSN Age: "clean sample" and "entrant sample"

SSN Age	Clean Sample	Entrant Sample
<19	16,455,525	5,805,684
19	182,481	53,950
20	155,933	55,257
21	155,743	57,114
22	149,210	51,894
23	153,427	52,314
24	157,685	50,705
25	156,152	46,924
26	150,045	41,481
27	140,610	33,525
28	130,243	28,040
29	119,788	22,399
30+	912,349	55,704

Table A2: Credit access, with additional geographic fixed effects

This table replicates Tables 2 and 4 with additional geographic fixed effects. The specification includes baseline fixed effects (birth year of the consumer, and consumers' first observed ZIP code, *First Zip5*), and adds last observed ZIP code (*Last Zip5*), longest observed ZIP code (*Longest Zip5*), and the number of unique ZIP codes observed (*Number Zip5*) as a measure of consumer mobility. The outcomes in Panel A are: age at first credit report (columns 1 and 2), age at first credit product (columns 3 and 4), age at first auto loan (columns 5 and 6), age at first mortgage (columns 7 and 8), and age at first credit card (columns 9 and 10). The outcomes in Panel B are: the probability of having ever held an auto loan by age 37 (columns 1 and 2), the probability of having ever held a mortgage by age 37 (columns 3 and 4), the probability of having ever held a credit card by age 37 (columns 5 and 6). The units in Panel A are years of age and the units in Panel B are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Panel A: Age of credit products first taken out

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15***	2.07***	4.73***	1.81***	3.88***	1.60***	1.72***	0.03	4.21***	1.49***
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)	(0.04)	(0.03)	(0.04)	(0.02)	(0.03)
SSN Age		0.72***		0.68***		0.53***		0.39***		0.63***
		(0.01)		(0.01)		(0.01)		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

Panel B: Probability of having held a credit product by age 37

Dep Var: By Age 37, has...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.61***	-5.62***	-5.21***	1.39***	0.09	0.37***
	(0.17)	(0.21)	(0.17)	(0.21)	(0.07)	(0.10)
SSN Age		-1.16***		-1.54***		-0.06***
		(0.04)		(0.04)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.137	0.137	0.151	0.151	0.070	0.070
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Table A3: Creditworthiness and credit card outcomes with additional geographic fixed effects

Panel A replicates Table 3 with additional geographic fixed effects. The specification includes baseline fixed effects (birth year of the consumer, and consumers' first observed ZIP code, *First Zip5*), and adds last observed ZIP code (*Last Zip5*), longest observed ZIP code (*Longest Zip5*), and the number of unique ZIP codes observed (*Number Zip5*) as a measure of consumer mobility. The units in Panel A columns 1 to 4 are credit score points, and the units in Panel A columns 5 to 8 are percentage points. Panel B uses the same specification. The outcomes in Panel B are: the number of credit cards at age 30 (columns 1 and 2) and at age 40 (columns 3 and 4); the dollar value of credit card limits at age 30 (columns 5 and 6) and at age 40 (columns 7 and 8). The units in columns 1 to 4 are number of credit cards, and in columns 5 to 8 are dollars. Standard errors are clustered by birth year × the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Panel A: Credit score outcomes

	Credit Score by...				Probability of Prime or Higher Score by...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	22.21***	14.48***	22.99***	12.59***	3.78***	6.05***	10.52***	6.33***
	(0.36)	(0.54)	(0.34)	(0.46)	(0.15)	(0.22)	(0.14)	(0.20)
SSN Age		1.91***		2.43***		-0.53***		0.98***
		(0.10)		(0.09)		(0.05)		(0.04)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.202	0.202	0.191	0.191	0.155	0.155	0.149	0.149
N	6,001,282	6,001,282	6,246,734	6,246,734	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	624.21	624.21	658.13	658.13	37.63	37.63	49.33	49.33

Panel B: Credit card outcomes

	Number of Credit Cards At...				Credit Card Limits At...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-0.13***	1.12***	0.57***	0.15***	-2,748.6***	3,027.2***	2,296.3***	1,539.3***
	(0.02)	(0.02)	(0.02)	(0.02)	(98.0)	(90.8)	(140.5)	(167.5)
SSN Age		-0.29***		0.09***		-1,345.2***		168.5***
		(0.00)		(0.00)		(21.0)		(32.0)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.126	0.128	0.096	0.096	0.141	0.143	0.182	0.182
N	6,293,372	6,293,372	4,914,884	4,914,884	6,293,372	6,293,372	4,914,884	4,914,884
Mean, SSN Age <21	3.35	3.35	3.92	3.92	$11,491.7	$11,491.7	$22,112.0	$22,112.0

Table A4: Credit outcomes by age 37, conditional on age 30 credit score

Odd-numbered columns in both panels are estimates from Equation 1; even-numbered columns are estimates from Equation 2. The outcome in Panel A is whether a consumer has any delinquency, measured by 90 or more days past due, by age 37, in percentage points. Each column of Panel A shows results for a different credit score group at age 30. Panel A, columns 1 and 2 show results for all credit scores. Panel A, columns 3 through 11 show results for the subsets of consumers with subprime (<600, the highest credit risk segment), nearprime (601-660), prime (661-720), prime plus (721-780), and superprime (781+, the lowest risk segment) credit scores. The outcome in Panel B columns 1 and 2 is the probability of having ever held an auto loan by age 37. In columns 3 and 4 it is the probability of having ever held a mortgage by age 37; in columns 5 and 6 it is the probability of having ever held a credit card by age 37. The units in columns 1 to 6 are percentage points. The outcome in columns 7 and 8 is the number of credit cards held. The units in columns 5 to 8 are dollars (credit card limits). Standard errors are clustered by birth year × the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Panel A: Any delinquency (90+ days past due) by age 37, split by credit score at age 30

Dep Var: Any delinquency by age 37	All		Subprime		Nearprime		Prime		Prime Plus		Superprime	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-1.50***	-1.23***	-2.64***	-3.31***	-3.51***	-2.65***	-1.44***	0.18	0.13	1.08***	0.29	0.42
	(0.25)	(0.39)	(0.40)	(0.74)	(0.35)	(0.54)	(0.25)	(0.48)	(0.10)	(0.17)	(0.15)	(0.29)
SSN Age		-0.07		0.18		-0.20*		-0.36***		-0.22***		-0.04
		(0.05)		(0.12)		(0.09)		(0.07)		(0.03)		(0.07)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X	X	X
R^2	0.185	0.185	0.032	0.032	0.028	0.028	0.027	0.027	0.025	0.025	0.022	0.022
N	5,804,866	5,804,866	2,538,267	2,538,267	905,644	905,644	881,507	881,507	1,008,294	1,008,294	471,154	471,154
Mean, SSN Age <21	27.18	27.18	45.45	45.45	26.17	26.17	13.11	13.11	4.52	4.52	1.63	1.63

Panel B: Credit outcomes by age 37

Dep Var: Outcomes at Age 37...	Any Auto Loan		Any Mortgage		Any Credit Card		Number of Credit Cards		Credit Card Limits	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-15.01***	-9.50***	-12.05***	-2.71***	-1.55***	-1.85***	0.36***	0.17***	-937.9	732.3
	(0.33)	(0.24)	(0.43)	(0.48)	(0.06)	(0.19)	(0.03)	(0.04)	(504.4)	(438.0)
SSN Age		-1.35***		-2.29***		0.07		0.05***		-409.3**
		(0.06)		(0.12)		(0.04)		(0.01)		(118.4)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X
R^2	0.108	0.108	0.267	0.268	0.111	0.111	0.141	0.141	0.305	0.305
N	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866	5,804,866
Mean, SSN Age <21	75.10	75.10	44.54	44.54	92.06	92.06	3.90	3.90	$20,431.9	$20,431.9

Table A5: Consumers still observed in 2024

Odd-numbered columns in both panels of this table are estimates from the cross-sectional OLS regression specified in Equation 1 that includes an indicator for immigration status (*Immigrant* is 1 if a consumer's SSN was assigned at age 21 or older). Even-numbered columns in both panels of this table are estimates from the OLS regression specified in Equation 2 that contains both the indicator for immigration status, and also *SSN Age*, the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group). All six regressions include fixed effects for the birth year of the consumer, fixed effects for consumers' first observed ZIP code (First Zip5), last observed ZIP code (Last Zip5), longest ZIP code (Longest Zip5) as a proxy for their most permanent location, and the number of unique ZIP codes (Number Zip5) as a proxy for their mobility. The outcome in columns 1 and 2 of Panel A is the probability of a consumer having a non-missing credit report in 2024. The outcome in columns 3 and 4 of Panel A is is the probability of a consumer having a non-missing credit score in 2024. The outcome in columns 5 and 6 of Panel A is is the probability of a consumer having a non-missing credit tradeline in 2024. Panel B replicates Panel B of Table 4 restricting the data to only the consumers with a non-missing credit score in 2024, and also including additional geographic fixed effects. The units are percentage points. Standard errors are clustered by birth year × the first two digits of a consumer's first observed ZIP code. *$p < .05$; **$p < .01$; ***$p < .005$.

Panel A: Probability of consumer still being observed in 2024

Dep Var: In 2024, has...	Any Credit Report		Any Credit Score		Any Credit Tradeline	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-2.10***	-4.38***	-6.25***	-11.19***	-7.39***	-11.83***
	(0.10)	(0.16)	(0.18)	(0.24)	(0.21)	(0.25)
SSN Age		0.53***		1.15***		1.04***
		(0.02)		(0.04)		(0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.045	0.045	0.112	0.113	0.147	0.147
N	6,293,372	6,293,372	6293372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	95.92	95.92	89.94	89.94	84.34	84.34

Panel B: Credit access by age 37, for consumers still observed in 2024

Dep Var: By Age 37, has...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.75***	-3.98***	-4.48***	4.88***	-0.49***	0.14
	(0.17)	(0.22)	(0.18)	(0.23)	(0.07)	(0.09)
SSN Age		-1.56***		-2.15***		-0.14***
		(0.04)		(0.05)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.11	0.111	0.138	0.138	0.061	0.061
N	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238
Mean, SSN Age <21	79.38	79.38	48.03	48.03	95.38	95.38

Table A6: American Community Survey: income, with additional fixed effects for employment status and occupation

This table uses public data from the 5-year American Community Survey (ACS) waves 2009, 2013, 2018, and 2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each column shows results of a separate cross-sectional regression. All columns include an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns also include *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. The outcome is Log (Income), the log of household income that is winsorized at 99.9^{th} percentile. The baseline mean is the household income in dollars for non-immigrants. All regressions also include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, survey year, employment status, and occupation. Units are percentage points. Heteroskedasticity-robust standard errors. $^{+}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .005$.

	Log (Income)							
	2006 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-11.23***	-7.57***	-10.98***	-6.50***	-9.15***	-5.43***	-4.41***	-1.27**
	(0.36)	(0.65)	(0.30)	(0.54)	(0.27)	(0.51)	(0.33)	(0.61)
Immigration Age		-0.93***		-1.05***		-0.79***		-0.66***
		(0.14)		(0.11)		(0.09)		(0.11)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
F.E. Employment	X	X	X	X	X	X	X	X
F.E. Occupation	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R^2	0.199	0.199	0.248	0.248	0.299	0.299	0.299	0.299
Mean Income, Immigration Age <21	$71,088	$71,088	$78,339	$78,339	$98,660	$98,660	$134,099	$134,099

Table A7: American Community Survey: access to mortgage, homeownership, and car ownership

This table uses public data from the 5-year American Community Survey (ACS) waves 2009, 2013, 2018, and 2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each column in each panel shows results of a separate cross-sectional regression. The outcome in Panel A is whether a consumer has any mortgage, the outcome in Panel B is whether a consumer is a homeowner (with or without a mortgage), and the outcome in Panel C is whether the consumer has any car. In all columns, the outcome is whether the consumer has any outstanding mortgage. All columns include an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older) and *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. All regressions also include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, and survey year. All columns except 1, 4, 7, and 10 (in all panels) include a control for Log (Income), the log of household income that is winsorized at 99.9^{th} percentile and is scaled to mean zero and standard deviation one. In all panels, columns 3, 6, 9, and 12 also include additional fixed effects for employment status, and occupation. Units are percentage points. Heteroskedasticity-robust standard errors. $^{+}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .005$.

	2006 - 2009			2009 - 2013			2014 - 2018			2019 - 2023		
A. Dep Var: Any Mortgage	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-9.90***	-7.57***	-6.90***	-8.09***	-6.21***	-5.38***	-3.56***	-2.09***	-1.23***	-1.86***	-1.40***	-0.65*
	(0.31)	(0.30)	(0.30)	(0.28)	(0.27)	(0.27)	(0.29)	(0.28)	(0.28)	(0.35)	(0.34)	(0.34)
Immigration Age	-1.98***	-1.69***	-1.73***	-1.63***	-1.32***	-1.34***	-1.36***	-1.14***	-1.13***	-0.76***	-0.60***	-0.56***
	(0.06)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Log (Income)		17.92***	18.13***		16.43***	16.41***		15.23***	14.88***		13.88***	13.12***
		(0.05)	(0.05)		(0.05)	(0.05)		(0.05)	(0.05)		(0.06)	(0.06)
R^2	0.100	0.213	0.220	0.116	0.205	0.212	0.135	0.206	0.213	0.131	0.192	0.199
Mean, Immigration Age <21	44.22	44.22	44.22	44.65	44.65	44.65	48.12	48.12	48.12	54.25	54.25	54.25
B. Dep Var: Homeowner	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-13.08***	-10.56***	-10.19***	-10.65***	-8.67***	-8.04***	-4.82***	-3.31***	-2.50***	-0.83**	-0.37	0.35
	(0.32)	(0.32)	(0.32)	(0.30)	(0.29)	(0.29)	(0.30)	(0.29)	(0.29)	(0.35)	(0.34)	(0.34)
Immigration Age	-2.02***	-1.70***	-1.79***	-1.75***	-1.42***	-1.50***	-1.56***	-1.34***	-1.38***	-1.10***	-0.93***	-0.94***
	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Log (Income)		19.37***	19.91***		17.31***	17.78***		15.62***	15.86***		13.80***	13.71***
		(0.05)	(0.05)		(0.05)	(0.05)		(0.05)	(0.05)		(0.06)	(0.06)
R^2	0.114	0.244	0.255	0.128	0.225	0.235	0.152	0.229	0.237	0.155	0.224	0.233
Mean, Immigration Age <21	52.40	52.40	52.40	53.25	53.25	53.25	59.26	59.26	59.26	68.54	68.54	68.54
C. Dep Var: Any Car	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-2.37***	-1.72***	-1.34***	-0.88***	-0.34*	0.00	1.11***	1.53***	1.69***	1.18***	1.30***	1.39***
	(0.25)	(0.24)	(0.24)	(0.20)	(0.19)	(0.19)	(0.16)	(0.16)	(0.16)	(0.17)	(0.17)	(0.17)
Immigration Age	-0.65***	-0.56***	-0.54***	-0.57***	-0.48***	-0.45***	-0.41***	-0.35***	-0.33***	-0.16***	-0.11***	-0.10***
	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Log (Income)		4.99***	4.85***		4.80***	4.61***		4.36***	4.15***		3.67***	3.48***
		(0.03)	(0.03)		(0.03)	(0.03)		(0.03)	(0.03)		(0.03)	(0.03)
R^2	0.200	0.234	0.236	0.211	0.240	0.243	0.187	0.215	0.219	0.153	0.177	0.181
Mean, Immigration Age <21	93.51	93.51	93.51	93.66	93.66	93.66	94.69	94.69	94.69	95.40	95.40	95.40
F.E. Demographics	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Employment & Occupation			X			X			X			X
N	2,079,500	2,079,500	2,079,500	2,162,012	2,162,012	2,162,012	2,231,275	2,231,275	2,231,275	2,200,018	2,200,018	2,200,018

Table A8: Survey of Consumer Finances (2022): debt attitudes and credit demand

This table uses public data from the 2022 Survey of Consumer Finances (SCF), keeping birth years 1972 to 2000 such that respondents are aged between 22 and 49. Odd-numbered columns in both panels are estimates from the cross-sectional OLS regression that includes an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns add *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. See Appendix A.2 for details of the survey variables used. The outcomes in Panel A are a variety of measures of attitudes to debt, as described more in Appendix A.2. The outcomes in Panel B are indicators for whether a consumer has applied for credit in the last twelve months: The outcome in column 1 is a binary variable for whether a consumer has applied for any credit in the past twelve months (constructed from variables listed below, and also variable x438 whether applied for student loan). Column 2 (constructed from variable x437) is whether applied for auto loan. Column 3 (constructed from variable x435) is whether applied for mortgage. Column 4 (constructed from variable x436) is whether applied to refinance mortgage. Column 5 (constructed from variable x433) is whether applied for credit card or respond to a pre-approved credit card. Column 6 (constructed from variable x434) is whether applied for a credit card limit increase. Column 7 (constructed from variable x439) is whether applied for other consumer credit. The units in both panels are percentage points. All regressions include a control for normal income (winsorized at the 99th percentile), and fixed effects for the birth year of the consumer and fixed effects for other demographic control variables: male, race, Hispanic, spouse, marital status, household size, number of children, and education levels. Heteroskedasticity-robust standard errors. Missing data in the SCF are imputed five times using a multiple imputation technique, storing data in five "implicates". We run a separate regression on each of the five implicates and follow the SCF's recommended multiply-imputed variance estimation technique for combining standard errors. R^2 is the adjusted r-squared averaged across the regressions for the five implicates. $^{+}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .005$.

Panel A: Debt attitudes

Dep Var: All Right To Borrow...	In General (Positive)		In General (Negative)		For Living Expenses		For Vacation		For Education	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-4.27	-4.29	4.69	11.65	-8.76^{+}	-8.35	-5.01	-8.37	-2.80	9.07
	(4.83)	(8.95)	(4.65)	(7.09)	(4.89)	(7.66)	(3.40)	(5.21)	(4.18)	(5.59)
Immigration Age		0.00		-0.89		-0.05		0.43		-1.52*
		(0.80)		(0.62)		(0.74)		(0.56)		(0.67)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.016	0.016	0.043	0.044	0.032	0.032	0.020	0.020	0.040	0.045
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1813
Mean, Immigration Age <21	29.51	29.51	25.71	25.71	73.38	73.38	15.30	15.30	84.24	84.24

Panel B: Credit demand

Dep Var: Apply For...	Any Credit		Auto		Mortgage		Credit Card		Other Credit	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32	-1.08	2.03	-6.55	2.11	5.65	1.63	2.02	1.84	-7.66^{+}
	(4.56)	(7.17)	(4.37)	(7.53)	(3.8)	(6.85)	(3.22)	(5.32)	(3.62)	(4.61)
Immigration Age		0.44		1.10		-0.45		-0.05		1.22*
		(0.62)		(0.72)		(0.65)		(0.48)		(0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08